A Bigger Learner Model Did Not Help

Geist Atlas · Module 7 · What the Tutor Needs to Know

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September 2026

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[**CLOSED**] Tested and closed under its stated gate · **Family:** What the Tutor Needs to Know
Claim: Detailed learner records, theory-of-mind additions, replay of alternative moves, fitted policies, and a logged hidden interior did not beat a lean baseline. An early advantage also disappeared after a turn-counting error was fixed: the gap was about 1.1× on both trap suites, within N≈24 noise.

A mostly negative arc with a useful by-product: strong trap scenarios and an auditable runner. The tutor did not need a richer written portrait of the learner; the testing apparatus was the durable result.

Derived view of `paper-full-2.0.md` v3.0.301 — §6.8, §6.9, §6.10. The paper is canonical; edit it there, not here.

6.8 Architectural Extension: The Adaptive Runner and Trap-Scenario Methodology

This section reports a second architectural-extension arc, parallel in structure to the id-director family in §6.7 but addressing a different gap. Where §6.7 asked whether charismatic-pedagogical authorship could be operationalised through a back-stage agent, §6.8 asks whether an externalised learner-state model and a programmatic policy graph can produce the *adaptive responsiveness* that §6.3 found absent under lightweight intervention and only partially recoverable through expensive search (§6.3.9). The extension is a heavier intervention than the best-of-N selector of §6.3.9: it replaces the dialogue engine's free-form turn loop with a LangGraph state machine, introduces an explicit

*This sentence is the only one actually authored by Liam Magee in this paper.

`learnerProfile` object that the tutor reads and writes across turns, and adds programmatic policy actions and counterfactual replay. Per §5.12.6, this section mixes reporting tiers: §6.8.3 reports a pre-registered confirmatory headline on `cell_110_langgraph_adaptive`; §6.8.4 reports the post-hoc exploratory dialogue-engine baseline (`cell_114`) together with the A13 within-LangGraph cells' binary scores on the same v1 suite (a re-classified exploratory ablation, per §5.12.2); §6.8.5 reports a pre-registered confirmatory P2.1 null on within-family discrimination; §6.8.6 reports the post-hoc exploratory P2.2 state-schema ablation (the "state-richness reversal" — a richer `learnerProfile` does not improve strategy selection; the two-field minimal profile `cell_118` scores highest on both the binary and graded instruments among the pre-registered cells, and the post-hoc `cell_123` ties it on the binary channel while sitting with the richer cluster on the graded one, localizing the binary floor to "any fourth structured field"); §6.8.7 reconciles the trap-suite result with the §6.3 adaptive-responsiveness null *and reports the post-hoc exploratory clean cross-suite test* (the trap-vs-suggestion-suite reconciliation referenced in §6.8.3 and §6.8.5 synthesises existing numbers; `cell_124`/`cell_125` on `config/cross-suite-trap-scenarios.yaml`, $N \approx 24$/cell, are new exploratory data discharging a forward-reference §6.8.3–§6.8.5 held open — the cross-architecture binary gap and the graded inversion both replicate the v1 suite); §6.8.8 is the closing synthesis — what the arc establishes, what it does not, and why the durable contribution is the apparatus rather than the architectures (synthesis, no new numbers); §6.8.9 adds a later post-hoc longitudinal character-state probe, which turns the local proof-DAG/resistance repair policy into a cross-scene learner-state layer and tests it with a generated synthetic learner. The pre-registered P2.3 crossover cells (`cell_121`, `cell_122`) are retired rather than deferred (§6.8.8): P2.1's null and P2.2's reversal have weakened the hypothesis they were built to probe, so running them on the original schedule would be sunk-cost behaviour. See `docs/explorations/claude/2026-05-10-p22-p23-parking-note.md` for the integration plan.

6.8.1 Motivation: Closing the §6.3.9 Gap Section 6.3.9 located *adaptive responsiveness* as structurally resistant to lightweight intervention: neither recognition prompts nor the multi-agent ego/superego architecture produced reliable turn-over-turn modulation of pedagogical strategy in response to learner-specific signals. Bridge 3 of that section established that best-of-N selection at $K = 3$ (~6× ego cost per turn) produces a small, statistically suggestive lift (Cohen's $d = +0.41$ on per-dialogue coupling) but does not breach the pre-registered "bridgeable" threshold. The §6.3.9 conclusion was that the gap is *partially* mitigable by expensive search but not closable by prompt-level or two-agent-level interventions.

The LangGraph adaptive runner introduced in this section is a substantially heavier intervention than best-of-N search. It crosses two design boundaries simultaneously. First, it externalises learner state out of the tutor prompt and into a structured `learnerProfile` object (fields: `confidence`, `misconceptions`, `agencySignal`, `zpdEstimate`, `lastEvidence`), updated by a dedicated graph node from each learner turn. Second, it replaces the dialogue engine's free-form turn loop with a programmatic policy graph: per turn the runner executes a fixed-order pipeline of nodes (context input, tutor ego, optional superego pass, profile update, optional policy action, optional constraint check, optional validator) determined by the cell's `adaptive.architecture` flag rather than by the tutor's free-text deliberation. The hypothesis the arc tests is whether either or both boundaries materially change adaptive-responsiveness performance, and whether the §6.3.9 gap is closable by an intervention of this weight.

The trap-scenario suite developed for this arc (described in §6.8.2) provides the corresponding measurement instrument: each scenario plants a hidden learner state (a specific misconception or behavioural signal) and a triggering turn at which an appropriately-adapted tutor would shift strategy. Because the trap is annotated, *whether* the tutor adapted is computable as a binary outcome (`strategy_shift_correctness`) and *how well* it adapted is scoreable on a 4-dimension graded rubric (§5.12.4). The two paths together permit the kind of mechanism-level claim §6.3 lacks: not merely "scores rise across turns" but "the right strategy shift fires at the right moment for the right hidden signal."

6.8.2 Method: Trap Scenarios, LangGraph Runner, and Dual Scoring Paths

Trap-scenario instrument. The trap suite (`config/adaptive-trap-scenarios.yaml`) consists of eight pedagogical situations in which a learner exhibits a *visible* surface behaviour that masks a different *hidden* underlying state. Each scenario specifies four fields: `hidden.actualMisconception` (the underlying state the tutor must diagnose), `hidden.triggerTurn` (the turn index at which the visible-vs-hidden mismatch becomes detectable), `hidden.triggerSignal` (the specific learner utterance that surfaces the mismatch), and `expectedStrategyShift` (the family of pedagogical moves an appropriately-adapted tutor would deploy at or shortly after the trigger). The eight scenarios span four pedagogical families — substantive engagement, scaffolding, diagnostic, and repair/affective — and are calibrated so that the trigger turn falls within the cell's 3-to-4-round dialogue budget. A v2 subset (`config/adaptive-trap-scenarios-v2.yaml`, six scenarios) preserves the trigger structure but constrains the expected shift to a single same-family alternative, supporting the within-family discrimination test reported in §6.8.5. The diagnostic structure of this instrument — recovering the hidden cause a surface utterance masks — is the multi-turn-tutoring analogue of *backtracing*, the task of retrieving the segment that caused a learner's query (Wang et al., 2024).

LangGraph adaptive runner. Cells configured with `runner: adaptive` in `config/tutor-agents.yaml` are dispatched to `services/adaptiveTutor/` rather than tutor-core's standard dialogue engine. The runner builds a state graph per the cell's `adaptive.architecture` flag. Five architectures are currently registered:

- `recognition_only`: contextInput → tutorEgo → done. A single LLM call per tutor turn; no profile update, no superego, no constraint check. Used as the within-LangGraph baseline (cells 111 and 111_v2 in §6.8.4 and §6.8.5).
- `ego_superego`: contextInput → tutorEgo → tutorSuperego → tutorEgoRevise → done. Two-pass tutor; still no profile update or constraint check (cell 112).
- `state_policy`: the full pipeline — contextInput → tutorEgo → learnerProfileUpdate → policyAction → constraintCheck → done (cell 110, the headline architecture in §6.8.3).
- `state_policy_with_validator`: adds an output-validator node after the constraint

check (cell 113).

- Bilateral and named-pattern variants: extend the profile-update node with an explicit ToM tracker or named-pattern lookup (cells 115–117 in §6.8.5).

The runner persists per-turn internal state — the `learnerProfile` snapshot, the policy action selected, the constraint-check result — to a JSON trace file under `logs/tutor-dialogues/` alongside the visible dialogue. Counterfactual replay is implemented as a deterministic re-execution of the same scenario under a perturbed `learnerProfile` (e.g., flipping `agencySignal` from `active` to `passive`) to surface what the tutor *would* have done under counterfactual learner state. Counterfactual divergence is reported descriptively in §6.8.5's descriptive-results paragraph and is not part of any pre-registered threshold.

Dual scoring paths. Adaptive dialogues are scored along two parallel paths because the v2.2 evaluator (§5.2.2) does not apply to them: `services/evalConfigLoader.js#getScenario()` reads only `config/suggestion-scenarios.yaml`, so trap-scenario dialogues return `SKIP (scenario not found)` from every v2.2 entrypoint. The first scoring path is *binary*: `scripts/analyze-strategy-shift.js` checks whether the tutor's response at or after `triggerTurn` matches the family specified by `expectedStrategyShift` (strict) or one of its same-family cousins (family-match). The second is *graded*: the bespoke 4-dimension rubric described in §5.12.4 (`trigger_recognition`, `strategy_execution`, `strategy_quality`, `pedagogical_coherence`, each 1–5) scores execution quality given that some action fired, with the non-blind and single-judge caveats made explicit there. The two paths are complementary: the binary asks *did the right family fire at the right time*; the graded asks *was the fire well-aimed and well-crafted*. Cells can lead on one path and lag on the other (cell_119 in §6.8.6's ablation table fires the right family more often than cell_110 but executes less crisply when it does).

Standard fan-out. Cells in this arc are run at $N \approx 24$–32 per cell, the exact fan-out varying by sub-study: the A13 v1 cells ran three runs × the eight v1 scenarios ($N \approx 24$); the v2 within-family cells (§6.8.5) ran four runs × the six v2 scenarios ($N = 24$); the P2.2 state-schema-ablation cells (§6.8.6) ran four runs × the eight v1 scenarios ($N = 32$); the post-hoc dialogue-engine baseline cell_114 (§6.8.4) ran three runs × the eight v1 scenarios ($N = 24$). Larger N appears for cells re-run across multiple weeks (e.g., cell_111 v1 accumulated $N = 71$ across A13 and follow-up fan-outs).

6.8.3 Headline: cell_110 Produces Measurable Strategy Shifts on the Trap Suite

The headline cell for the architecture is `cell_110_langgraph_adaptive` — the full `state_policy` LangGraph configuration with externalised profile, programmatic policy actions, and constraint check. The pre-registered confirmatory result on the v1 trap-scenario suite (run id `eval-2026-05-05-486d7d1e`, $N = 23$):

Metric	Value	Notes
`strict_shift_correctness`	47.8% (11/23)	The tutor's at-or-post-trigger response matches `expectedStrategyShift` family exactly
`family_match_rate`	87.0% (20/23)	The response is in the same pedagogical family as expected, even when not the exact target
Graded `trigger_recognition` (mean)	4.26 / 5	Tutor identified the trap signal at or near `triggerTurn`
Graded `strategy_execution` (mean)	4.30 / 5	Post-trigger action aligned with `expectedStrategyShift`
Graded `strategy_quality` (mean)	3.96 / 5	Execution was specific, calibrated, non-leaky
Graded `pedagogical_coherence` (mean)	3.61 / 5	Whole trajectory coheres as teaching
Graded overall (mean of 4)	4.03 / 5	—

The result is moderate rather than strong. The strict signal hits 47.8% — the tutor selects the exact pre-registered family in roughly half of trap-trigger cases — and the family-match signal hits 87.0%, indicating the tutor is in the right pedagogical neighbourhood almost always. The graded path shows the architecture's profile: trigger recognition and strategy execution are strong (both above 4.25), but pedagogical coherence is the weakest dimension (3.61), suggesting that the full-state architecture picks the right move but produces somewhat bloated or muddied trajectories around it. The binary 47.8% provides the first measurement showing an architectural intervention produces detectable strategy shifts on a pre-registered trap-trigger instrument — a different result from §6.3.9's lightweight-intervention-resistant finding, but on a different stimulus suite (see §6.8.7 on the cross-suite comparison). Whether this lift represents a genuine architectural advance over the dialogue engine or simply a within-LangGraph baseline is the question §6.8.4 and §6.8.7 address.

6.8.4 The Dialogue-Engine Baseline on the v1 Trap Suite (`cell_114`, Post-Hoc)

Per §5.12.2, none of the A13 cells (110–113) actually ran the dialogue engine — all four are LangGraph sub-graphs. `cell_114_dialogue_engine_trap_baseline` (§5.12.3) closes that gap: tutor-core's base single-agent ego prompt, driven turn by turn on the v1 trap suite (8 scenarios $\times$ 3 runs $= N = 24$, fanned out across 4 parallel lanes of 2 scenarios each; run ids `eval-2026-05-12-c53a1d8f` / `-f157bcd0` / `-3db6bf3e` / `-41f4487f`), with the same trap-steered `learnerTurn` learner and the same `claude-code/sonnet` model the A13 cells use. It is post-hoc and exploratory (§5.12.6); the cell_111 cell_114 contrast is confounded with prompt-assembly path (§5.12.3), so the result is a *directional dialogue-engine floor*, not a controlled single-factor isolation.

The binary instrument (`scripts/analyze-strategy-shift.js`), applied identically to all five cells on the v1 suite (A13 v1 runs `eval-2026-05-05-486d7d1e` / `-b3ac505b` / `-e0e615bc` / `-e58b7da1`):

Cell	Architecture	`strict_shift`	`window` (pivot anywhere post-trigger)	`family_match`
cell_110_langgraph_adaptive	full `state_policy`	47.8% (11/23)	73.9% (17/23)	87.0% (20/23)
cell_113_a13_C4_validator	`state_policy` + validator	41.7% (10/24)	58.3% (14/24)	66.7% (16/24)
cell_111_a13_C1_recognition_only	`recognition_only`	37.5% (9/24)	58.3% (14/24)	50.0% (12/24)
cell_112_a13_C2_egosuperego	`ego_superego`	30.4% (7/23)	60.9% (14/23)	39.1% (9/23)
cell_114_dialogue_engine_trap_baseline	**dialogue engine (base ego)**	**25.0% (6/24)**	**25.0% (6/24)**	**58.3% (14/24)**

The dialogue engine is the floor on strategy selection — and, on the stored convention, *appears* uniquely never to re-pivot (largely a scoring artifact, corrected at the end of this subsection). Its `strict_shift` of 25.0% is the lowest of the five cells; every LangGraph variant beats it, including the weakest (cell_112 ego_superego, 30.4%), and the full state-policy cell beats it by ~23 pp. More diagnostic than the level is the gap between `strict` and `window`: all four LangGraph cells pivot to the expected family on a *later* post-trigger turn in a meaningful share of cases (window above strict by +17 to +30 pp across the four), whereas cell_114's window rate is *identical* to its strict rate. Once the base ego picks a move at trigger+1 it stays in that family for the rest of the dialogue. This is the behavioural signature of having no externalised learner-state representation and no constraint check: the dialogue engine processes each turn as "respond to the latest message", with nothing tracking "the learner just sprang a trap; check whether my move landed". [*Substantially corrected below* (turn-convention correction, end of §6.8.4): the `strict` = `window` identity for cell_114 — the basis for "never re-pivots" — is largely an artifact of a turn-index misalignment in the baseline adapter; event-anchored, the engine reaches the expected family within two post-trigger turns ~45% of the time.]

The family-confusion matrix shows the base ego's prior. Nineteen of cell_114's 24 trigger+1 moves fall in the broad `substantive_engagement` family (`mirror_and_extend`, `acknowledge_and_redirect`, and the engagement-style moves `scope_test` / `name_the_disagreement` / `pose_counterexample`) — a strong "keep elaborating the content" default. The consequence is sharp on the trap families that require *detecting a hidden learner state* rather than adding content: cell_114 scores **0/6** on `repair_affective` traps (`affective_shutdown`, `repair_after_misrecognition` — where the learner has disengaged or been misread and the right move is to slow down, de-escalate, or explicitly repair) and **1/3** on the `diagnostic` trap (`polite_false_mastery` — where the right move is to probe the learner's hidden reasoning). cell_110, by contrast, hits 4/6 on `repair_affective` and

3/3 on `diagnostic`: its externalised profile flags the affective shift and its programmatic policy menu makes the de-escalation/repair move available and salient. The base ego is *not incapable* of these moves — it plays `scope_test` on the argue-back traps (`false_confusion`, `resistance_to_insight`: 2/3 each, where engaging substantively *is* the trap-appropriate response, so its default happens to be right), `ask_diagnostic_question` on one `polite_false_mastery`, and even `repair_misrecognition` in 2 of 3 `repair_after_misrecognition` dialogues — but it fires those moves inconsistently and without timing discipline. In the two `repair_after_misrecognition` dialogues where the base ego *does* play `repair_misrecognition`, it does so at the trigger turn itself (sandwiched between `mirror_and_extend` moves on either side), not at the trigger+1 slot the binary scores, so the instrument records 0/3 for that scenario type even though the move is present. The mid-pack `family_match` (58.3%, above cell_111 and cell_112, below cell_110 and cell_113) is the same story read the other way: the broad `substantive_engagement` family is easy to land on the 12 trap scenarios where some form of substantive engagement is the target (12/12 family-correct there), and it is the exact-label/timing discipline (`strict`, `window`) plus the non-content families (`repair_affective` 0/6) where the dialogue engine bottoms out.

The graded rubric inverts the ranking — and that inversion is the finding. Under the 4-dimension adaptive graded rubric (§5.12.4; GPT-5 via the codex CLI bridge, $N = 24$), cell_114's overall mean is **4.46/5** — *higher* than cell_110's 4.03 ($N = 23$, same judge):

Cell	`trigger_recognition`	`strategy_execution`	`strategy_quality`	`pedagogical_coherence`	Overall (mean of 4)
cell_110_langgraph_adaptive	4.26	4.30	3.96	3.61	4.03
cell_114_dialogue_engine_trap_baseline	**4.54**	**4.21**	**4.50**	**4.58**	**4.46**

A reader who looked only at the graded rubric would conclude cell_114 is the better tutor. This makes the §5.12.4-limitation-1 distinction concrete. The graded rubric scores *quality of execution given that some action fired* — fluency, calibration, coherence of the resulting prose — and a mature, heavily-iterated single-agent tutor prompt produces fluent, coherent teaching even when it selects the wrong strategic family; cell_110's externalised-state machinery, which buys it the correct move more often on the stored-convention binary (≈1.9×, but convention-fragile — see the turn-convention correction at the end of this subsection), also bloats and muddies its trajectories (its weakest graded dimension, 3.61 on coherence, is §6.8.3's own observation, and the cell_114 gap is widest exactly there: 4.58 vs 3.61). The two instruments measure different constructs — *which move, when* (binary) versus *how well-crafted the prose* (graded) — and the architectural intervention trades the second for the first. This is not a "GPT-5 rewards verbosity in cell_114's favour" artefact: the dialogue-engine turns are *long* (400–2700 characters), the opposite of the concision GPT-5 is documented to prefer (§5.12.4 limitation 2), so the high graded score reflects genuine prose coherence, not a length bias. The one place binary and graded converge is `strategy_execution`, the single dimension built to ask about post-trigger family alignment: there cell_114 (4.21) and cell_110 (4.30) are roughly tied, and cell_114's `strategy_execution` distribution carries the spread the binary predicts — one row at 1 (the `false_confusion` dialogue that mirrored when `scope_test` was the target), five at 3 (defensible-but-not-target moves), against a body of 4s and 5s. The graded rubric *can* see the selection problem, but only on the dimension designed to look for it; the other three reward the prose and outvote it.

Bottom line. On the v1 trap suite, the standard dialogue engine — same model, same trap-steered learner, same scenarios as cell_111 — does not produce the strategy shifts the LangGraph state-policy configuration produces: it sits at the bottom of the binary `strict_shift` ranking (25.0% vs 30–48%), it uniquely fails to re-pivot on later turns, and it collapses entirely on the trap families that require detecting a hidden learner state rather than elaborating content. The most plausible reading is that externalised learner-state tracking plus a constraint check are what let the LangGraph cells pivot — but the cell_111 cell_114 contrast also varies the prompt-assembly path (§5.12.3), so this is a directional baseline floor rather than a clean single-factor result — and, per the turn-convention correction that immediately follows, the binary gap it rests on is itself an artifact: the secure re-run under the fixed adapter (§6.8.4) puts both legs at ~1.1× (v1 1.15×, cross-suite 1.07×), within $N \approx 24$ noise — no detectable advantage. The graded rubric inverting the ranking (4.46 > 4.03) is itself the result: graded prose quality and strategic-move correctness are not the same construct, and the dialogue engine wins the first while losing the second. Per §5.12.6: post-hoc exploratory; no alpha correction; single judge throughout (binary `claude-code/sonnet`; graded GPT-5/codex); no blinding.

Turn-convention correction (v3.0.119): the cross-architecture binary gap is convention-fragile, and the "never re-pivots" signature is largely a scoring artifact. The `repair_after_misrecognition` observation three paragraphs up — the base ego plays the correct repair move "at the trigger turn itself … not at the trigger+1 slot the binary scores," so the instrument records 0/3 "even though the move is present" — was the visible symptom of a *systematic* scoring artifact, not a tutor timing deficit. `scripts/run-dialogue-engine-trap-baseline.js` advanced the learner-turn index by one (`turn: turn + 1`) where the LangGraph runner passes `turn: state.turn`; because the trap learner fires its trigger on the turn index it is handed (`realLLM LEARNER_TURN_SYSTEM`: "if this is the trigger turn, surface the signal"), the dialogue-engine learner sprang the trap one tutor-turn *early*, and the fixed-index scorer (which reads `triggerTurn + 1`) therefore read the engine's *second* post-trigger response, not its first. The offset is architecture-correlated — the LangGraph cells pass `state.turn` and are aligned; only the dialogue-engine baseline is shifted — so it inflated the cross-architecture gap rather than adding symmetric noise. The convention was centralized and the adapters fixed in `scripts/lib/trapTurnConvention.js` (commit `3474ebf`). An offline, event-anchored re-score (`scripts/rescore-trap-strict-shift.js`; `exports/trap-strict-shift-rescore.json`) re-produces the published numbers exactly under the stored convention, then re-scores by locating each trigger's actual emission in the transcript (by its signal text) and scoring the first tutor turn after it.

The corrected reading is a *grid*, because the gap depends on two independently-defensible scoring choices — where the trigger is located (fixed `triggerTurn + 1` vs event-anchored) and how strict the match is (exact label vs pedagogical family):

convention	v1: cell_110 vs cell_114	cross-suite: cell_124 vs cell_125
strict · stored (as published above)	47.8% vs 25.0% (1.91×)	62.5% vs 30.4% (2.05×)
strict · event-anchored	43.5% vs 25.0% (1.74×)	62.5% vs 43.5% (1.44×)
family · stored	87.0% vs 58.3% (1.49×)	66.7% vs 39.1% (1.70×)
family · event-anchored	65.2% vs 66.7% (0.98×)	66.7% vs 47.8% (1.39×)

Each correction independently deflates the gap, and they compound. The locator-free `family · stored` row — which uses the *published* fixed offset, no re-location — already cuts the v1 advantage from 1.91× to 1.49× on label-granularity alone (the engine's `request_elaboration` / `mirror_and_extend` moves sit in the same *family* as the targeted `ask_diagnostic_question` / `scope_test`). Event-anchored location cuts it again, to 0.98× (v1) and ~1.4× (cross-suite) under the most defensible offline convention — enough to show the gap was *not secure*, but the offline event-anchored locator fights noise on the old mis-aligned transcripts, so a clean re-run was needed to settle the number. The "uniquely never re-pivots" claim above is the same artifact read through the `window` metric: event-anchored, cell_114 reaches the expected family within the first two post-trigger turns in ~45% of cases (`win 2` in the artifact), well above its 25.0% stored `window` rate — it *does* re-pivot; the scorer was reading past the pivot. This *adds to* rather than replaces the §5.12.3 prompt-scaffold confound, which still applies on top.

The secure re-run (v3.0.121) settles it. Cells 114 and 125 were re-run under the fixed adapter (fresh sample, $N = 24$ each, `claude-code/sonnet`; run ids `eval-2026-06-06-460fc561` and `-5a168684`) and scored by the now-aligned instrument (fixed adapter + `triggerTurn + 1` scorer agree by construction, so no event-anchoring is needed). The engine floor, previously undercounted by the artifact, rises sharply: `cell_114` (v1) `strict_shift` 25.0% → **41.7%** (10/24), family 58.3% → **75.0%**; `cell_125` (cross-suite) `strict_shift` 30.4% → **58.3%** (14/24), family 39.1% → **62.5%**. (Mechanism-level confirmation: `cell_114`'s `repair_affective` hits at trigger+1 go from 0 to 10 — the affective traps it was being silently mis-scored on.) Against the frozen state cells (`cell_110` 47.8% / 87.0%; `cell_124` 62.5% / 66.7%) the secure contrasts are **v1 1.15× strict / 1.16× family** and **cross-suite 1.07× strict / 1.07× family** — gaps of ~4–6 pp, well within $N \approx 24$ sampling noise (no test would call a ~5-pp gap significant at this N). This lands slightly *above* the grid's most-aggressive corner (the fresh aligned run is authoritative over the noisy offline locator). One caveat, stated not hidden: the state cells are frozen (old epoch), the engine cells fresh, so this is not a same-epoch contrast — but the artifact was engine-adapter-only and the state cells were always aligned, and the conclusion is robust to the caveat because the gaps are small either way (re-running `cell_110`/`cell_124` was judged not worth the ~47 paid dialogues). **Final secure verdict, replacing "the correct move twice as often, the dialogue engine never re-pivots": once the turn**

convention is fixed, the cross-architecture binary advantage is ~1.1× and not a detectable difference on either suite — the published ~2× was the artifact. Power caveat — this is *not-detected*, not *demonstrated-zero*. At $N \approx 24$/cell the comparison is severely underpowered for an effect this small: the pooled secure gap is +5.3 pp, 95% CI $[-15, +25]$ pp (Fisher $p = 0.68$; per-suite v1 $[-22, +35]$, cross-suite $[-23, +32]$), and detecting a true 5-pp difference at 80% power would need ~1,400 dialogues per cell. A small real advantage (~1.1×, *consistent in direction* across both suites and with the state machine's explicit re-pivot, which the locally-greedy engine lacks) is therefore **not excluded** — what the re-run removes is the *~2× magnitude* (which rested on the engine's mis-measurement at 25%, vs its clean 41.7%, CI $[24, 61]$%), not the *existence* of a small effect. The residual is recorded as *not-detected, not absent*, and is not worth ~1,400 paid dialogues per cell to chase, given that any such edge is binary-channel-only (the graded rubric inverts) and remains scaffold-confounded (§5.12.3). Status (§5.12.6): post-hoc correction of a post-hoc exploratory result; single judge throughout (`claude-code/sonnet`); artifacts `exports/trap-strict-shift-rescore.json` (offline grid) plus the two re-run ids above. This correction governs the cross-suite replication (§6.8.7) and the §6.8.8 synthesis equally.

6.8.5 Bilateral-ToM Elaboration Adds No Measurable Within-Family Discrimination (P2.1) The pre-registered P2.1 fan-out (`docs/explorations/claude/2026-05-10-p2-followup-pr` §P2.1, locked 2026-05-07; runs 2026-05-08 to 2026-05-10) tested whether elaborating the LangGraph `learnerProfile` update with an explicit bilateral Theory-of-Mind (ToM) tracker — a node that maintains separate first-person and third-person learner-state representations — produces within-family discrimination beyond the `recognition_only` baseline. Four cells were run at $N = 24$ each on the v2 trap-scenario suite (six within-family scenarios where every valid alternative belongs to the same pedagogical family as the target):

Cell	Architecture	`strict_shift_correctness`
cell_111_a13_C1_ recognition_only_v2	`recognition_only` baseline	33.3% (8/24)
cell_116_recognition_named_ patterns_v2	`recognition_named_patterns` (named-pattern lookup added)	37.5% (9/24)
cell_115_bilateral_tom_v2	`bilateral_tom` (first/third-person ToM tracker)	45.8% (11/24)
cell_117_bilateral_tom_ named_patterns_v2	`bilateral_tom_named_ patterns` (both)	33.3% (8/24)

The H1.1 primary contrast pools by architecture family:

Pooled architecture	shift_n / n	shift%
bilateral_tom (cells 115 + 117)	19 / 48	39.58%
recognition_only (cells 111 + 116)	17 / 48	35.42%

The pooled gap is **+4.17 percentage points** in favour of bilateral_tom, with a 95% Wald CI of [−15.18, +23.52] pp. The pre-registration locked binary point-estimate thresholds (lines 30–34, 70–72 of the pre-reg doc): a gap of ≥ 10 pp supports H1.1 (bilateral_tom adds within-family discrimination), 5–10 pp is inconclusive, and < 5 pp falsifies H1.1 in favour of H1.3 (the v1 family-match advantage of bilateral_tom was a between-family, type-recognition effect rather than a within-family discrimination effect). The observed +4.17 pp falls below the 5-pp falsification threshold. **H1.3 is supported as written; the bilateral_tom elaboration does not produce within-family discrimination at the pre-registered effect size.** The CI width permits a true 10-pp gap that sampling chance compressed to 4 pp, but the pre-reg authors locked point-estimate thresholds knowingly, and this report applies them as written.

The H1.2 secondary endpoint (`family_match_rate` ≥ 80% predicted; < 70% falsifying) corroborates the H1.3 reading. Three of the four cells fall below the 70% falsification floor on v2: cell_116 (66.67%), cell_115 (66.67%), and cell_117 (58.33%); only cell_111 (70.83%) escapes, by 0.83 pp. All three falsified cells had v1 family-match > 70% (cell_115 = 70.8%, cell_116 = 79.2%, cell_117 = 79.2% on the v1 suite per the upstream A13 follow-up memo), so the v2 drop is interpretable per the pre-reg's rule: the cells' v1 family-match advantage was scenario-distribution-driven (v1 scenarios spanned three pedagogical families and made between-family discrimination easy), not architectural (v2 scenarios constrain to a single family and the architectures collapse).

Descriptive results not part of the pre-reg's locked thresholds are reported for completeness. The four cells handle `substantive_engagement` scenarios well (cells 111, 115, 116 all score 12/12 family-correct, cell_117 scores 9/12) but collapse on `diagnostic_opaque_reasoning` (0–1 of 4 family-correct across all four cells); scaffolding falls in between (3–5 of 8 across cells). Per-scenario, only two of six scenarios drive most of the strict_shift signal: `substantive_correct_underspecified` (10/16 across the four cells, 62.5%) and `scaffolding_load_not_gap` (14/16, 87.5%); the other four scenarios sit at ≤ 25% strict. The pre-reg forbids post-hoc subsetting (line 84); these per-scenario numbers are reported as descriptive context rather than as a re-analysis. Counterfactual divergence on the two ToM cells (cell_115 cf_div 62.5%, cf_fam 37.5%; cell_117 cf_div 62.5%, cf_fam 54.2%) is consistent with type-prior locking in cell_115 but inverted in cell_117.

Two limitations apply to P2.1 alongside the §5.12.6 reporting convention. First, single-judge scoring: all four cells were scored under `claude-code/sonnet`; a second-judge probe was not pre-registered and has not been run. Second, the wide CI on the H1.1 contrast (39 pp wide, $[-15, +24]$ pp) means the locked binary rule is more decisive than the underlying inferential evidence: a future $N = 48$ replication per cell would tighten the CI by approximately $\sqrt{2}$, sufficient to distinguish a true null from a true 10-pp effect.

The pre-reg's "Order of execution" section (line 239) flagged that a P2.1 null would weaken the prior on P2.2 (state-schema ablation) and P2.3 (bilateral × id-director crossover) without invalidating them, since P2.2's diagnostic value for the cosmetic-vs-substantive question is partly independent of whether the architecture does within-family work, and P2.3's three-endpoint design (charisma + accuracy + strategy_shift) is only partially dependent on the strategy_shift endpoint — §6.8.8 records the subsequent decision to retire P2.3 rather than run it on the original schedule. The interpretation that follows in §6.8.6 (state-richness reversal) and §6.8.7 (cross-architecture status) applies the pre-reg's framework: the architecture appears to recognise pedagogical *type* but not to discriminate *within* a type when valid alternatives are same-family, and the project's claim space narrows accordingly.

6.8.6 State-Schema Ablations: A Richer Learner Profile Does Not Help (the State-Richness Reversal) The headline `cell_110` (§6.8.3) carries a structured `learnerProfile` object with five fields the policy reads each turn — `confidence` (scalar), `misconceptions` (array), `agencySignal` (5-state enum), `zpdEstimate` (string), `lastEvidence` (a quoted learner utterance). Is that structure doing work, or is it cosmetic dressing on what is really a `confidence`-scalar-plus-dialogue-text policy? The P2.2 sub-study (pre-registration §P2.2; cooling period closed 2026-05-09) answers this with three ablation cells that project the LLM-emitted profile down to a subset of fields before the policy sees it — the model still infers the full profile each turn, so inference cost is held constant; only the policy's *view* is narrowed. **H2.1** predicted a monotone dose-response on `strict_shift`: `cell_110` (full) $\geq$ `cell_119` (drops `misconceptions`) $\geq$ `cell_120` (drops `agencySignal`) $\geq$ `cell_118` (`confidence` + `lastEvidence` only), with `cell_110 - cell_118` $\geq$ `15` pp; **H2.2** predicted `cell_120` would drop $\geq$ 15 pp on the repair/affective family (the agency enum was theorised to govern affect-tracking) while holding on substantive scenarios; **H2.3**, the falsifying alternative, was that the structured fields beyond `confidence + lastEvidence` are cosmetic. Each ablation cell ran four runs $\times$ the eight v1 trap scenarios ($N = 32$); `cell_110`'s comparison numbers are its A13 v1 fan-out ($N = 23$).

Cell	`learnerProfile` fields the policy sees	N	`strict_shift`	Graded overall (T / E / Q / C)
`cell_110` (`state_policy`, full)	confidence, misconceptions, agencySignal, zpdEstimate, lastEvidence	23	47.8%	4.03 (4.26 / 4.30 / 3.96 / 3.61)
`cell_120` (`no_agency_signal`)	confidence, misconceptions, zpdEstimate, lastEvidence	32	50.0%	4.09 (4.25 / 3.81 / 4.22 / 4.09)
`cell_119` (`no_misconceptions`)	confidence, agencySignal, zpdEstimate, lastEvidence	32	53.1%	4.06 (4.09 / 3.78 / 4.25 / 4.13)
`cell_118` (`minimal_profile`)	confidence, lastEvidence	32	68.8%	4.34 (4.34 / 4.28 / 4.41 / 4.34)
`cell_123` (`minimal_plus_zpd`, post-hoc)	confidence, lastEvidence, zpdEstimate	32	68.8%	4.11 (4.03 / 4.00 / 4.28 / 4.13)

(T / E / Q / C = `trigger_recognition` / `strategy_execution` / `strategy_quality` / `pedagogical_coherence`, each 1–5; graded overall is their mean; binary `strict_shift` = correct pre-registered family at trigger+1; both scored as in §6.8.3, binary under `claude-code/sonnet`, graded under GPT-5/codex.)

H2.1's predicted ordering is reversed — and the reversal is a step, not a slope. The observed `strict_shift` ordering is `cell_118` (68.8%) ≥ `cell_119` (53.1%) ≥ `cell_120` (50.0%) ≥ `cell_110` (47.8%) — the reverse of the predicted order at every comparison that exceeds sampling noise. `cell_110 - cell_118 = -21` pp against a predicted ≥ `+15` pp: opposite sign, larger magnitude than the predicted effect. All three of H2.1's predicted positive gaps over `cell_118` come out negative (`cell_110` −21 pp, `cell_120` −19 pp, `cell_119` −16 pp). But the reversal is carried *entirely* by `cell_118`: the three richer cells — `cell_110`, `cell_119`, `cell_120` — form a flat cluster at 47.8 / 53.1 / 50.0% (spread 5.3 pp, comfortably inside the ±10-pp band that $N \approx$ 24–32 Bernoulli sampling produces around a true ~0.5), so their internal order (predicted `cell_110` ≥ `cell_119` ≥ `cell_120`, observed `cell_119` ≥ `cell_120` ≥ `cell_110`) is uninterpretable noise and is not interpreted. `cell_118`'s lead over that cluster — the one comparison the reversal actually rests on — is itself near the edge of binary-channel significance at $N = 32$: ~18 pp over the pooled three-cell mean, bootstrap 95% CI ≈ $[-1, +37]$ pp, label-permutation $p \approx 0.10$. It is firmer on the graded channel (a +0.28-point overall gap, CI ≈ $[+0.04, +0.51]$, $p \approx 0.01$), it is broad rather than scenario-bound (`cell_118` matches or beats the cluster on six of the eight trap types — the lone two it does not, `resistance_to_insight_v1` and `sophistication_upgrade_v1`, are scenarios *every* cell fails), and §5.12.4's cross-judge calibration reproduces the graded ordering under Gemini. So the direction *minimal* ≥ *rich* is carried by the graded instrument plus that scenario-level breadth plus the judge replication, not by the 21-pp binary headline alone. The structure is in the *discontinuity*, and it points the wrong way: collapsing the profile to two fields lifts `strict_shift` by ~19 pp over the full five-field object; pulling any single mid-tier field (`misconceptions`, `agencySignal`) does nothing. H2.3's prediction — "the fields beyond `confidence + lastEvidence` are cosmetic" — is supported in spirit and slightly exceeded in fact: they are not merely inert, they are mildly counterproductive on this instrument; the externalised-state machine's measured advantage over recognition-only (`cell_111`, §6.8.4) is doing its work through `confidence` plus the quoted-dialogue anchor, not through `misconceptions` / `agencySignal` / `zpdEstimate`. (The pre-reg's literal falsification rule — `cell_110 - cell_118` differ by `< 5` pp → cosmetic — does not mechanically fire, since `|-21|` is not `< 5`; but a 21-pp reversal is the cosmetic verdict and then some, and per the pre-reg's "mixed outcomes are reported as such" instruction it is reported as a reversal, not patched into a re-ordered hypothesis.)

H2.2 is unsupported, also in the wrong direction. `cell_120` was built to test whether the agency enum drives repair/affective discrimination. It scored 7/8 `strict_shift` on the repair/affective traps (`affective_shutdown_v1` 4/4, `repair_after_misrecognition_v1` 3/4) — at or above `cell_110`'s 4/6 — i.e., removing `agencySignal` did *not* depress the family it was theorised to govern, and `cell_120` held on substantive scenarios besides. Whatever lets the policy recognise repair/affective situations, it is not the structured agency enum; the live candidates are `lastEvidence`'s quoted utterance and the ego LLM's own reading of the transcript. (`n = 6`–8 per cell on this sub-family, so the apparent 21-pp *gain* is itself within noise; the safe statement is "no drop", which is enough to retire H2.2.)

Graded scoring agrees within the pre-registered cells — but `cell_123` is a divergence. Unlike the cross-*architecture* comparison of §6.8.4, where the binary and graded instruments diverge (the dialogue engine writes more coherent prose, picks the wrong family more often — by a convention-fragile margin, §6.8.4), within the P2.2 family's *four pre-registered cells* the two instruments *agree*: `cell_118` leads on both (graded 4.34, every dimension ≥ 4.28, and a near-perfect 96.9% family-match), and `cell_110` / `cell_119` / `cell_120` cluster on both (graded 4.03 / 4.06 / 4.09). §5.12.4's independent-judge calibration confirms the `cell_118` > `cell_119` > `cell_110` graded ordering holds under both GPT-5 and Gemini, so it is not a single-judge artefact; `cell_120`, graded only after that calibration run, is single-judge GPT-5 and slots between `cell_118` and `cell_119`, statistically tied with `cell_119` and `cell_110`. The lone within-P2.2 binary/graded *divergence* is the post-hoc `cell_123` — top on binary (= `cell_118`), mid on graded ($\approx$ the ~50% cluster) — examined in the next paragraph. One sub-pattern is GPT-5-only: the two four-field cells (`cell_119`, `cell_120`) carry a depressed-`strategy_execution` left tail — roughly a third of rows scored   3, versus ~1/20 for `cell_110` and `cell_118` — which §5.12.4 reports the Gemini re-judge cannot resolve (Gemini's ceiling effect collapses the tail), so it stays a GPT-5 observation rather than a finding.

What `cell_123` settles — `zpdEstimate` is a free rider on move selection; the binary destabiliser is the *fourth structured field*, not a specific one. The P2.2 design pulls one field at a time off the *full* profile, so it left three readings of the floor open: "`zpdEstimate` is the destabilising field" (it is the only structured field present in all three ~50% cells and absent from the ~69% `cell_118`), "having ≥ 4 structured fields at all", and "any one of `{misconceptions, agencySignal, zpdEstimate}`". The post-hoc cell `cell_123` (`state_policy_minimal_plus_zpd`: `{confidence, lastEvidence, zpdEstimate}` — `cell_118`'s set plus `zpdEstimate`, behaviourally an exact clone of `cell_118` with one more field projected through to the policy; $N = 32$, v1 suite) adjudicates: it scores `strict_shift` **68.8% — the same 22/32 as `cell_118`**, with a 100% family-match (every trigger+1 move in the correct broad family) and a per-scenario profile recognisably the same shape as `cell_118`'s (4/4 on `answer_seeking` / `affective_shutdown` / `repair_after_misrecognition`, 0/4 on the universally-hard `false_confusion`). So **`zpdEstimate` is exonerated**: the two cells that differ *only* in `zpdEstimate` (`cell_118`, `cell_123`) are tied to the row, so adding it to the lean `{confidence, lastEvidence}` core costs nothing — which kills the first reading and locates the floor in the *other* structured fields. Nor is it a single toxic field among those: `cell_119` (keeps `agencySignal`, drops `misconceptions`) = 53.1% and `cell_120` (keeps `misconceptions`, drops `agencySignal`) = 50.0% both sit in the ~50% cluster — each cell is missing the field the other reading would blame, and is still depressed, so neither "`misconceptions` is toxic" nor "`agencySignal` is toxic" survives. What moves the policy from the ~69% lean cluster (`cell_118`, `cell_123`) to the ~50% rich cluster (`cell_119`, `cell_120`, `cell_110`) is *the presence of either `misconceptions` or `agencySignal`* — equivalently, crossing from three to four structured fields — and the effect *saturates*: `cell_110`, carrying both, is 47.8%, not detectably below the one-each cells, so the second destabilising field adds essentially nothing. Saturation is the signature of a load/capacity ceiling (the policy LLM does single-decision strategy selection worse once it must integrate four-plus structured fields) rather than additive per-field toxicity. The honest residual: a 3-field cell carrying `{confidence, lastEvidence, misconceptions}` (or `…, agencySignal`) would separate "pure field-count" from "specifically that field" — ~69% under a count story, ~50% under a content story — but it has not been run and is not planned; the data in hand favour the count/load reading via the saturation point without excluding the content one. Either way, on the binary channel the §6.8.4 externalised-state advantage over recognition-only runs through `confidence`, the quoted-dialogue anchor `lastEvidence`, and (harmlessly) `zpdEstimate` — not through the `misconceptions` array or the `agencySignal` enum.

One caveat: `zpdEstimate` is free on *move selection*, not on *execution prose*. On the *graded* channel `cell_123` does **not** join `cell_118`: its GPT-5/codex overall is 4.11 (4.03 / 4.00 / 4.28 / 4.13) — squarely in the ~50% cluster's graded band (4.03–4.09), well below `cell_118`'s 4.34 — and it carries the same depressed-`strategy_execution` left tail as the four-field cells (9/32 rows scored 3, against ~1–2/32 for `cell_118` and `cell_110`). So `zpdEstimate` is a free rider on *which move, when* (the binary `strict_shift` channel, where adding it to the lean core costs nothing) but a modest tax on *how well the resulting teaching is executed* (the graded channel, where adding it drops `cell_123` to the rich-cluster level): the graded "minimal wins" result of the pre-registered cells does not extend to `cell_123`, which is the lone within-P2.2 case where the binary and graded instruments diverge (cf. the cross-*architecture* divergence of §6.8.4). Per §5.12.4 the `strategy_execution` left tail is a GPT-5-only signal — Gemini's ceiling effect cannot resolve it, and `cell_123` was graded after that calibration run, so its graded position carries the same single-judge caveat as `cell_120`'s; the binary result (`cell_123` = `cell_118` to the row) is the more robust half of the cell.

Status and caveats (per §5.12.6). Cells 118–120 are pre-registered confirmatory tests of H2.1–H2.3 (pre-reg §P2.2); `cell_123` is a post-hoc exploratory follow-up, not pre-registered. Binary `strict_shift` is single-judge `claude-code/sonnet`; graded scoring is single-judge GPT-5/codex with the §5.12.4 Gemini inter-rater check on cells 110/118/119 only (`cell_120` and `cell_123`, graded after that calibration run, are GPT-5-only — and `cell_123`'s graded position, including the `strategy_execution` left tail it shares with the four-field cells, carries the §5.12.4 GPT-5-only caveat). No alpha correction is applied across the four (then five) cells; the `strict_shift` gaps among `cell_110` / `cell_119` / `cell_120` are within sampling noise and not interpreted, and `cell_118`'s lead over that cluster is firm on the graded channel (permutation $p \approx 0.01$) but only marginal on the binary `strict_shift` channel (permutation $p \approx 0.10$, bootstrap CI brushing zero) — the reversal's *direction* is treated as established, the 21-pp binary figure as the high end of a wide interval. Counterfactual replay (`counterfactual.enabled: true` in the configs) is a no-op for the P2.2 architectures — they fall outside the runner's profile-update set, so `counterfactual_total = 0` for every P2.2 cell — and the comparison rests entirely on the on-path binary and graded metrics. N: `cell_110` = 23 (its A13 v1 fan-out); cells 118 / 119 / 120 / 123 = 32 (four runs × eight v1 scenarios). This result narrows the project's "externalised cognition" claim space exactly as the pre-reg's H2.3 anticipated, and §6.8.7 already folds it into the cross-suite reconciliation (the trap-suite gain is between-*type*, not within-type, and is not a graded-quality effect).

6.8.7 Reconciling the Trap-Suite Result with the §6.3 Adaptive-Responsiveness Null §6.3.8 reported adaptive responsiveness as null under every experimental factor tested (recognition prompts, multi-agent architecture, learner architecture: all slope effects below the $d = 0.27$ detectability threshold — pooled $d = 0.03$ for recognition, ≤ 0.15 for the architecture factors — across $N = 432$ dialogues), and §6.3.9 reported the insight-action gap as "structural under prompt design and lightweight architectural intervention; partially mitigable only by expensive generation-space search" — Bridges 0–2 land within ± 0.01 of the cell-40 baseline coupling, and Bridge 3 (best-of-K=3, ~6× ego cost) reaches $d = 0.41$, suggestive but below the pre-registered bridgeable threshold. §6.8.3–§6.8.4 then reported that the LangGraph `state_policy` runner produces detectable strategy shifts on the v1 trap suite (`strict_shift` 47.8% for cell_110 versus 25.0% for the dialogue-engine baseline cell_114). Read carelessly, this looks like a contradiction: an architectural intervention *did* move adaptive behaviour. It is not a contradiction. The §6.3 null and the trap-suite positive are about different outcome variables, on different stimuli, at different intervention weights — and the second is consistent with, indeed forecast by, the first.

Different outcome variable; different instrument sensitivity. The §6.3 null is on a *diffuse, aggregate* signal: per-dialogue rubric slopes across free-form multi-turn dialogues (§6.3.2, §6.3.5), and the reflection-action coupling cosine within turn (§6.3.9). "Did the tutor adapt" there is inferred from whether scores trend upward across turns or whether introspective text resembles delivered text — both are low-resolution proxies, and both are exactly the kind of measure that washes out a single localized strategic move inside an otherwise-unchanged trajectory. The trap suite is a *localized, pre-annotated, single-decision* instrument: each scenario plants a hidden state and a trigger turn, and `strict_shift` asks one question — did the pre-registered family of moves fire at trigger+1. A binary check on a designed decision point is a far more sensitive detector of "the tutor selected the right strategic move" than a slope on free-form dialogue, the same way a planted bug is a more sensitive test of a linter than waiting for bugs to surface in production. The §6.3 instruments were built to ask whether *recognition* steepens adaptation *on average over a dialogue*; the trap instrument asks whether *an architecture* selects *the right move at one annotated moment*. A null on the first and a positive on the second are jointly coherent: the architectural effect is real but it lives in a channel — localized strategic-move selection — that the §6.3 measures were not designed to resolve.

Different intervention weight; §6.3.9's own escape clause names this class. §6.3 tested *lightweight* manipulations — prompt orientation, a second (superego) agent, learner-architecture variants. §6.3.9 added a cost ladder (explicit instruction → two-pass reflection injection → retargeted critique → best-of-K=3) and found only the heaviest rung crossed the suggestive threshold, with its own forward-looking caveat: closing the gap would require "much higher K in best-of-K search, *or genuinely new architectures with explicit memory and iterative refinement*". The LangGraph `state_policy` runner is the second of those: it externalises learner state into a structured `learnerProfile` object that persists and is rewritten across turns (explicit memory), and it replaces the free-form turn loop with a programmatic policy graph plus a constraint check (a structured refinement of the move, not just the prose). The trap-suite positive is therefore best read not as a contradiction of §6.3 but as a partial *confirmation* of §6.3.9's hypothesis: lightweight interventions are inert, expensive search is suggestive, and a heavy externalised-state architecture is the first thing in the project to produce a measurable shift on a sensitive instrument. “Measurable” is doing bounded work — 47.8% strict is moderate, not strong (§6.8.3), and the dialogue-engine comparison carries the §5.12.3 prompt-scaffold confound that makes cell_114 a *directional floor* rather than a clean single-factor baseline (§6.8.4) — but moderate-and-confounded is still a different result from §6.3's clean null, and the difference sits exactly where §6.3.9 said it would.

What the trap-suite positive does *not* license. Four bounds keep it consistent with the rest of §6:

- *Not a recognition-prompt effect.* §6.3's actual null is about recognition modulating adaptation; the §6.8 cells do not isolate recognition-on versus recognition-off within a fixed runner — cell_111 `recognition_only` is itself a LangGraph sub-graph (§6.8.2), not a prompt manipulation of the dialogue engine — so nothing here revives the recognition-modulates-adaptation hypothesis. The effect is architectural, not orientational.
- *Not a within-type effect.* §6.8.5's pre-registered P2.1 test found the architecture adds no within-family discrimination (pooled +4.2 pp, below the 5-pp falsification floor): it appears to recognise pedagogical *type* — substantive versus scaffolding versus diagnostic versus repair/affective — but not to choose well *among same-type alternatives*. The trap-suite gain is a between-type gain.
- *Not a graded-quality effect.* The instruments that do *not* isolate the single strategic decision — the §6.3 trajectory slopes, the §6.3.9 reflection cosine, and the trap suite's own 4-dimension graded rubric (which scores whole-trajectory prose quality) — find the architecture null or *inferior*: §6.8.4's graded rubric inverts the cell ranking, with the dialogue engine ahead of cell_110 on prose quality, calibration, and coherence (4.46 versus 4.03) even as it picks the wrong family more often (the §6.8.4 binary gap, now known to be convention-fragile). The one instrument that *does* isolate that decision — the trap binary `strict_shift` — is the one that puts the architecture ahead. The architectural advantage is specifically an advantage in *which move, when*, not in *how well-crafted the resulting teaching*.
- *Transfers across instruments, in one direction, with the same confound.* The clean cross-suite test now exists — a trap-style annotated instrument re-derived from six §6.3 suggestion scenarios (`config/cross-suite-trap-scenarios.yaml`), run on both architectures (`cell_124` = `cell_110`'s `state_policy` runner, `cell_125` = `cell_114`'s dialogue engine), scored with the same binary `strict_shift` — and it reproduces the v1 result: `cell_124` 62.5% (15/24) versus `cell_125` 30.4% (7/23), a ratio (≈2.05×) within rounding of the v1 suite's (1.91×) — both on the *stored* convention; the §6.8.4 turn-convention correction shows this gap is an artifact, and the secure re-run under the fixed adapter puts both legs at ~1.1× (cross-suite 1.07×, v1 1.15×), within $N \approx 24$ noise (no detectable advantage) — with the graded rubric again inverting (`cell_125` 4.60 > `cell_124` 4.22). Details and bounds in the “Discharging the forward-references” paragraph below: the claim now holds on two independently-built trap instruments, but it remains a *directional* comparison (`cell_125` carries the same §5.12.3 prompt-scaffold confound as `cell_114`), runs *one direction only* (no one has run the dialogue engine *with* the LangGraph runner's prompt scaffold), is *between-type* (the cross-suite

gap concentrates in the affective-shutdown and productive-deadlock scenarios; on two scenarios — `misconception_surfaces`, `activity_avoidance` — *neither* architecture fires the annotated move), and lives *on the binary channel* (the dialogue engine writes more coherent prose on both suites).

The picture across both suites, then, is internally consistent rather than contradictory:

Stimulus suite	Instrument	What it isolates	Finding
Suggestion (§6.3, $N = 432$)	Per-dialogue rubric slope	Adaptation rate over a whole dialogue, by condition	Null ($d \leq 0.03$ pooled; §6.3.2, §6.3.8)
Suggestion (cells 40–45 + bridges)	Reflection-action coupling cosine	Whether introspective noticing reaches the delivered message	Structural; only best-of-K suggestive ($d = 0.41$; §6.3.9)
Trap v1 ($N \approx 24$/cell)	`strict_shift` (binary)	Right family at trigger+1	cell_110 47.8% vs cell_114 25.0% *stored* → artifact; secure re-run 47.8% vs 41.7% = 1.15×, n.s. (§6.8.4)
Trap v1 ($N \approx 24$/cell)	4-dimension graded rubric	Prose quality given some action fired	*Inverts*: cell_114 4.46 > cell_110 4.03 (§6.8.4)
Trap v2 ($N = 24$/cell)	`strict_shift`, within-family	Right move among same-type alternatives	Null (pooled +4.2 pp, sub-threshold; §6.8.5)
Cross-suite (trap-annotated §6.3 subset, $N \approx 24$/cell)	`strict_shift` (binary)	Right family at trigger+1, on §6.3-derived scenarios	cell_124 62.5% vs cell_125 30.4% *stored* (≈2.05×) → artifact; secure re-run 62.5% vs 58.3% = 1.07×, n.s. (§6.8.4)
Cross-suite ($N \approx 24$/cell)	4-dimension graded rubric	Prose quality given some action fired	*Inverts*, as on v1: cell_125 4.60 > cell_124 4.22 (§6.8.7)

Discharging the forward-references. §6.8.3 flagged two open questions for §6.8.7 and §6.8.4 to settle, and §6.8.5 reserved §6.8.7 for the "cross-architecture status" reading. *Cross-architecture status:* §6.8.4 answered the within-LangGraph half — the dialogue-engine baseline sits below all four LangGraph cells on `strict_shift` (a gap §6.8.4's turn-convention correction shows to be convention-fragile, with the "uniquely never re-pivots" signature largely a turn-misalignment artifact) — so the cell_110 lift is not a within-LangGraph artefact, subject to the §5.12.3 prompt-scaffold confound that keeps it a directional floor rather than a clean isolation. *Cross-suite reconciliation:* the trap-suite positive and the §6.3 null are not in tension, for the reasons above — different instruments, different constructs, different intervention weights, with the trap positive consistent with §6.3.9's own forecast.

The clean cross-suite test, now run. The genuinely outstanding item §6.8.3–§6.8.5 reserved

— run the LangGraph `state_policy` runner and the dialogue-engine baseline on a trap-style-annotated subset of the §6.3 suggestion scenarios, scored with the same binary `strict_shift`, so that "the heavy architecture beats the dialogue engine on localized strategic-move selection" can be checked off the trap suite it was measured on — has now been executed. `config/cross-suite-trap-scenarios.yaml` re-derives six trap scenarios from §6.3's suggestion suite (`epistemic_resistance_impasse`, `affective_shutdown_impasse`, `productive_deadlock_impasse`, `misconception_correction_flow`, `activity_avoider`, `struggling_learner`), spanning all four policy families (two substantive-engagement, one diagnostic, one repair/affective, two scaffolding), in the same hidden-state-plus-trigger format the v1 suite uses — so both runners and both scorers consume it unchanged. Two new cells run it at $N \approx 24$ each: `cell_124_langgraph_adaptive_crosssuite` (cell_110's `state_policy` graph, including counterfactual replay; 24/24 dialogues) and `cell_125_dialogue_engine_crosssuite_baseline` (cell_114's standard single-agent ego via the same `run-dialogue-engine-trap-baseline.js` adapter; 23/24, one transient generation failure). The result reproduces the v1 pattern in every respect:

- *Binary, same direction, same ratio.* `cell_124` selects the pre-registered family at trigger+1 on 62.5% of scenarios (15/24; family-match 66.7%) versus `cell_125`'s 30.4% (7/23; family-match 39.1%) — a ≈2.05× advantage, within rounding of the v1 suite's 1.91× (47.8% vs 25.0%; §6.8.3–§6.8.4). Both absolute levels run higher on the §6.3-derived scenarios than on the bespoke v1 traps, plausibly because trigger signals re-derived from the well-worn suggestion suite are less ambiguous; the *contrast* between architectures is essentially the same magnitude. As on v1, the stored convention records every hit at trigger+1 with `cell_125` (like `cell_114`) never re-pivoting — but, per the §6.8.4 turn-convention correction, that "never re-pivots" signature is largely a turn-misalignment artifact of the dialogue-engine adapter (event-anchored, the engine does re-pivot within the post-trigger window), so it should not be read as a clean architectural property.
- *Graded, same inversion.* The 4-dimension graded rubric again flips the ranking: `cell_125` 4.60 versus `cell_124` 4.22 (mean over trigger-recognition, strategy-execution, strategy-quality, pedagogical-coherence), with the widest single-dimension gap on coherence (4.78 vs 4.08) — the dialogue engine writes the more coherent, better-calibrated trajectory even as it picks the wrong family more often (a stored-convention margin; convention-fragile per §6.8.4). `cell_124`'s own profile mirrors cell_110's: high trigger-recognition (4.58 — it *notices* the moment) but its weakest dimension is strategy-execution (3.96 — the prose carrying out the chosen move is rougher than the dialogue engine's). The split is, as in §6.8.4, real: the architectural advantage is in *which move, when*, not in *how well-crafted the teaching*.
- *Between-type, with shared blind spots.* The cross-suite gain concentrates in two scenarios — `cross_affective_shutdown` (`cell_124` 4/4 vs `cell_125` 1/4: the dialogue engine reads the affective-shutdown trigger as a cue to keep delivering content rather than to acknowledge-and-redirect) and `cross_productive_deadlock` (3/4 vs 1/4) — with a smaller edge on `cross_epistemic_resistance` (4/4 vs 3/4). On `cross_misconception_surfaces` *neither* architecture fires the annotated `ask_diagnostic_question` (both 0/4 — both prefer to engage the surfacing misconception substantively rather than diagnose

it, which may say as much about the annotation as about the tutors), and on `cross_activity_avoidance` both are weak (`cell_124` 1/4, `cell_125` 0/4). The advantage is a between-type gain — the architecture more reliably recognises *which kind* of move the moment calls for — exactly as §6.8.5's within-family null implies it should be.

So the cross-architecture claim now holds on **two** independently-constructed trap instruments rather than one. It remains, on both, a *directional* comparison: `cell_125` carries the same §5.12.3 prompt-scaffold confound as `cell_114` (its ego prompt is tutor-core's base ego, not the adaptive runner's `tutorEgoInitial` builder), so the contrast bundles "no LangGraph state machinery" with "different prompt scaffold"; and it runs one direction only — no one has run the dialogue engine *with* the LangGraph runner's prompt scaffold, the symmetric isolation that would separate the two. With that caveat held, the §6.8.4 bottom line now reads: on two trap suites — the bespoke v1 and the §6.3-derived cross-suite — with their respective trap-steered learners, the `state_policy` architecture selects the pre-registered strategic move more often than the dialogue engine on the stored-convention binary (≈1.9–2.05×), while the dialogue engine writes more coherent prose; the effect is between-type, confined to the binary channel, confounded by the prompt scaffold, replicated rather than single-suite, and — per the §6.8.4 turn-convention correction — mostly an artifact: the secure re-run under the fixed adapter puts both legs at ~1.1× (cross-suite 1.07×, v1 1.15×), within $N \approx 24$ noise, so the durable claim is *no detectable binary advantage* on either suite, not the "twice" magnitude. The §6.3 results are unchanged by any of this. Per §5.12.6: the §6.8.7 reconciliation of the §6.3 null synthesises existing numbers, but the cross-suite cells (`cell_124`, `cell_125`) are *new post-hoc exploratory data* — not in the A13 pre-registration, status exploratory, no alpha correction, single-judge throughout (binary scoring `claude-code/sonnet`; graded scoring GPT-5/codex), no blinding — carried here because they discharge a forward-reference §6.8.3–§6.8.5 explicitly held open. Every comparison in this section that involves cell_114 or cell_125 is post-hoc exploratory; the §6.8.3 and §6.8.5 results it cites carry their own (confirmatory, pre-registered) status.

6.8.8 Closing Synthesis: The Apparatus Is the Contribution; the Elaborations Mostly Are Not Stepping back from the cell-by-cell results to the question §6.8 opened with — did an externalised learner-state model plus a programmatic policy graph deliver the adaptive responsiveness §6.3 found absent? — the answer is *partially: on the trap binary, replicated across two independently-built trap suites, in a comparison still confounded by prompt scaffold, and not through the mechanisms the design assumed.* That answer, read alongside §6.7's parallel verdict on the id-director family, locates the real contribution of these two architectural-extension arcs: not in the architectures, but in the apparatus built to test them (the general point is developed in §7.4).

What is defensible. Four findings survive their caveats:

1. *The trap-scenario methodology works.* Externally-defined `expectedStrategyShift` annotations, a hidden-state-conditioned learner that springs a trigger at an annotated turn, counterfactual replay, and a binary `strict_shift` correctness check together turn "did the tutor adapt" — diffuse and slope-shaped in §6.3 — into a localized, scoreable event. That methodology is a contribution independent of any architecture it was used

to evaluate; §6.3.9's account of why the slope-and-cosine instruments wash out a single strategic move is, in effect, the negative-space argument for why an instrument like the trap suite was needed (§6.8.7).

2. *The `state_policy` architecture produces a moderate strategy-shift advantage over the dialogue-engine floor, replicated on a second trap suite.* On the v1 trap suite, `cell_110` selects the pre-registered family of moves at trigger+1 more often than the dialogue-engine baseline `cell_114` on the stored-convention binary (47.8% vs 25.0% `strict_shift`; §6.8.3–§6.8.4) — though, per the §6.8.4 turn-convention correction, that gap is convention-fragile and the apparent "the dialogue engine never re-pivots" property is largely a turn-misalignment artifact (the v1 advantage does not survive event-anchored family scoring). The clean cross-suite test (§6.8.7) reproduces this on a trap instrument re-derived independently from the §6.3 suggestion scenarios: `cell_124` (cell_110's architecture) 62.5% vs `cell_125` (cell_114's architecture) 30.4% — a ≈2.05× advantage, within rounding of the v1 ratio (1.91×) — with the graded rubric inverting on the cross-suite exactly as it does on v1 (`cell_125` 4.60 > `cell_124` 4.22). This is the first result in the project where an architectural intervention — not a prompt, not a second agent, not a learner-architecture swap — moves a sensitive adaptive-behaviour measure, and it now moves it on two suites rather than one. But it is *mostly an artifact* (the secure re-run under the fixed adapter puts both legs at ~1.1× — v1 1.15×, cross-suite 1.07× — within $N \approx 24$ noise, i.e. no detectable binary advantage; the §6.8.4 turn-convention correction), and what little may remain is *between-type* not within-type (§6.8.5), *confined to the binary channel* (the graded rubric inverts the ranking on both suites — `cell_114` 4.46 > `cell_110` 4.03; `cell_125` 4.60 > `cell_124` 4.22; §6.8.4, §6.8.7), *confounded* by the prompt-scaffold difference that makes `cell_114`/`cell_125` directional floors rather than clean single-factor baselines (§5.12.3), and *not isolated in the symmetric direction* — the dialogue engine has never been run *with* the LangGraph runner's prompt scaffold, so "architecture, not scaffold" remains an inference rather than a measured contrast on either suite.

3. *Within the state machine, less learner state is more.* The P2.2 ablation (§6.8.6) reversed its own pre-registered dose-response prediction: the two-field minimal profile `cell_118` scores highest on `strict_shift` (68.8%) and on the graded rubric (4.34); the three richer cells (`cell_110`/`cell_119`/`cell_120`) form a flat ~50% cluster; and the post-hoc `cell_123` localizes the binary floor to "any fourth structured field" with a load/capacity signature (the effect saturates), not a single toxic field. The structured `learnerProfile` the architecture was built around does its measurable work, on this instrument, through `confidence` and a quoted-dialogue anchor — the named-`misconceptions` array and the fixed-vocabulary `agencySignal` enum are mildly counterproductive, and `zpdEstimate` is close to inert (a modest tax on graded execution prose, free on move selection).

4. *Bilateral-ToM elaboration adds nothing measurable.* The P2.1 test (§6.8.5) was a clean pre-registered null — no within-family discrimination over `recognition_only` (pooled +4.2 pp, below the 5-pp falsification floor). Negative, but the cleanest-status finding in the section. Read against the broader Theory-of-Mind benchmark literature, this null and the §6.8.6 state-richness reversal are plausibly *regime*-bounded rather than

instrument defects: GroupToM-Bench (Tang et al., 2026) locates the payoff of explicit social-mental-state modelling at the *group* level, where collective outcomes emerge non-linearly from conformity and structural pressure and cannot be recovered by summing individual intents (its "linear-superposition-bias" failure mode) — a regime our strictly *dyadic*, single-learner trap suite never enters, so the inertness here of explicit first/third-person ToM and a structured mental-state schema is consistent with that machinery becoming load-bearing only once more than two parties interact. Extending the apparatus to group / >2-party interaction dynamics — where that structure is expected to start paying — is a forward direction this arc flags but does not test.

What is not established. Three gaps bound the arc:

- *No clean cross-architecture claim, though the cross-suite gap is now replicated.* The dialogue-engine comparison (`cell_114`, and its cross-suite twin `cell_125`) bundles "no LangGraph state machinery" with "different prompt scaffold" (§5.12.3), so it is a directional floor, not an isolation; the only other cross-architecture data point is a thin $N = 6$ id-director overlap (`cell_106`, never re-run at adequate N). The "no trap-style instrument for the §6.3 suite" gap is closed: §6.8.7's `config/cross-suite-trap-scenarios.yaml` re-derives six trap scenarios from the suggestion suite, and on it `cell_124` beats `cell_125` 62.5% vs 30.4% on `strict_shift` *on the stored convention* (≈2.05×) — but the secure re-run under the fixed adapter collapses this to 62.5% vs 58.3% — i.e. 1.07× — within $N \approx 24$ noise (§6.8.4) — with the graded inversion also replicating; so the cross-architecture *direction* (state ≥ engine) holds weakly on two independent trap instruments, but the *magnitude* is an artifact: corrected, there is **no detectable binary advantage** on either suite (v1 1.15×, cross-suite 1.07×, both n.s.; §6.8.4). What is still missing is the *symmetric* isolation — running the dialogue engine *with* the LangGraph runner's prompt scaffold — which is what would turn "architecture, not scaffold" from an inference into a measured single-factor contrast on either suite.
- *The hypothesised mechanisms mostly are not doing the hypothesised work.* The design's three architectural bets were rich externalised learner state, programmatic policy actions, and counterfactual replay. P2.1 falsified the bilateral elaboration; P2.2 reversed the state-richness prediction and pinned the residual advantage on two scalar/text fields; counterfactual replay produced only a messy signal where it ran (high raw divergence but low family-coherence on P2.1's two ToM cells — §6.8.5) and was a literal no-op for the P2.2 ablations (`counterfactual_total = 0` for every P2.2 cell, the ablation architectures falling outside the runner's profile-update set). Individually each is a finding; collectively they say the elaborations the harness made testable did not behave as the architecture assumed.
- *P2.3 (the crossover cells) is retired, not deferred.* The pre-registered fan-out scheduled two further cells (`cell_121`, `cell_122`) for a P2.3 crossover test; P2.1's null and P2.2's reversal have substantially weakened the hypothesis those cells were built to probe, so running them on the original schedule would be sunk-cost behaviour and the arc closes here rather than carrying an open P2.3 tail. (If a crossover question is revived it should follow a fresh decision about what it now answers, not the original plan.)

Relation to §6.3.9 and §6.7. This arc does not reverse §6.3.9's central finding — the insight-action gap is structural under lightweight intervention and only suggestively mitigable by expensive search. It *extends* it: a heavy externalised-state architecture is the first thing in the project to produce a measurable shift on an instrument sensitive enough to see one, but the shift is moderate, channel-specific, confounded, and — the part that matters for the mechanism story — it *survives stripping the architecture down to a two-field profile*, which is not what a "rich learner model drives adaptation" hypothesis predicts. Read with §6.7, the pattern rhymes: the id-director family found charismatic-pedagogical authorship could be operationalised but that mechanism stacks sit *inside* the Pareto frontier rather than pushing it out (§6.7's synthesis); the adaptive-runner family found adaptive strategy selection could be operationalised but that the architecture's elaborations mostly did not pay. In both arcs the durable result is the apparatus — the id-construction trace and the charisma rubric in §6.7, the trap suite and the dual scoring paths here — and the architectural elaborations the apparatus made legible came back null, negative, or inverted more often than not. That is a finding about how hard it is to make these elaborations pay, reported as such, in the spirit of §5.12.6's instruction that mixed and negative outcomes are reported rather than patched into re-ordered hypotheses.

Status (per §5.12.6). §6.8.8 introduces no new numbers; it synthesises §6.8.1–§6.8.7 (themselves a mix of pre-registered confirmatory — §6.8.3, §6.8.5 — and post-hoc exploratory — §6.8.4, §6.8.6, and §6.8.7's `cell_124`/`cell_125` cross-suite cells — tiers, with §6.8.7's §6.3-null reconciliation a no-new-data synthesis layered over those new exploratory cells). No alpha correction applies to a synthesis. The retirement of P2.3 is a scope decision, recorded here so the pre-registered fan-out's open items are accounted for rather than silently dropped.

6.8.9 Postscript: From Local Repair to Longitudinal Character-State Routing A later DAG/resistance follow-up tested a narrower question than §6.8's trap-suite architecture comparison: once the proof-DAG policy and resistance-signal router can repair a local scene, can their evidence be carried forward as a compact learner *character state* that reduces repeated repair on later scenes? The implementation adds a six-axis state ledger (`proof_ownership`, `relevance_orientation`, `frustration_tolerance`, `question_consolidation`, `non_formulaic_reasoning`, `next_step_agency`) updated only from observed outcome evidence, then runs the same learner identity through six linked scenes (`boredom`, `frustration`, `irrelevance`, `question_flood`, `rote_parroting`, and a transfer scene). The decisive contrast holds the v2 local policy fixed and varies only whether the carried state is routed into the generated learner context: `v2_policy_only` versus `character_state_plus_v2`. The learner sees sanitized character-state summaries and prior scene outcomes, not the target evidence labels; the closed-loop tutor policy remains programmatic. Implementation and artifacts: `services/adaptiveTutor/characterState.js`, `scripts/run-dag-resistance-character-development.js`, and `exports/adaptive-dag-resistance-charact`

On a generated synthetic-learner run using the real learner backend (`codex` provider, `model=auto`, three seeds $\times$ six scenes $= N = 18$ per arm), the state-routed arm improves efficiency over an already strong v2 baseline: `v2_policy_only` succeeds on 17/18 scenes, with 13/18 first-response successes and 4/18 staged follow-ups; `character_state_plus_v2` succeeds on 18/18 scenes, with 15/18 first-response successes and 3/18 staged follow-ups. Both arms hit 3/3 first-response successes on the transfer scene. The mock generated-learner contrast is lower and sharper (`v2_policy_only` 0/18 successes, `character_state_plus_v2` 6/18), confirming that the scaffold can create a developmental signature when the learner does not spontaneously provide evidence, but the real generated learner is sufficiently competent that the added state layer buys a modest efficiency gain rather than a qualitative rescue.

The claim is therefore bounded. This is **not** human character development, and it is not evidence that the tutor's philosophical-recognition orientation produces longitudinal learner transformation. It is a post-hoc synthetic mechanism probe showing that local adaptation evidence can be accumulated into a cross-scene state representation and routed back into later learner uptake without exposing answer labels, yielding a small improvement in first-response closure under real generated learner calls. In the terminology of §6.8.8, the contribution is again apparatus-first: the policy layer can now distinguish "repair succeeded locally" from "the learner has become more self-directing across scenes," and can measure the latter with first-response success, staged-follow-up rate, and transfer closure. Whether the same state variables track durable human learner development remains a pilot question, not a result here.

A second follow-up folds the same state ledger into a synthetic **Character-DAG drama** benchmark: a fixture-driven arc (`setup`, `pressure`, `peripeteia`, `consolidation`, `transfer`) couples proof-DAG/resistance policy to dramatic pressure and character-state routing, with a shuffled-state arm as the negative control. The strict real generated-learner robustness run uses three decisive arms, three seeds, eight scenes, and four fixture perturbations (`baseline`, `noisy_openings`, `harder_transfer`, `state_dependent_transfer`; $N = 24$/arm/perturbation). `full_character_dag_drama` passes every perturbation: first-response success is 21/24 in all four slices, compared with `policy_only` at 16/24, 12/24, 17/24, and 11/24, and the shuffled-state control at 14/24, 11/24, 9/24, and 9/24. The transfer channel is the sharpest contrast: full reaches 9/9, 8/9, 9/9, and 9/9 first-response transfer closure; `policy_only` reaches 8/9, 5/9, 9/9, and 1/9; shuffled-state reaches 4/9, 4/9, 3/9, and 2/9. The harder-transfer perturbation exposes a policy-ceiling case (`policy_only` and full both 9/9 transfer), so the gate is recorded as "full transfer stronger or policy ceiling matched" while the robustness layer still requires an aggregate full-policy transfer margin across the strict set. In all four perturbations, full keeps the staged-followup-or-unresolved burden at 3 scenes, observes 3/3 required peripeteia events with zero unrequired detections, and records zero target-label and public theory/process leaks. Implementation and artifacts: `config/character-dag-drama-framework.yaml`, `scripts/run-character-dag-drama-framework.js`, `scripts/run-character-dag-drama-robustness.js`, `services/adaptiveTutor/outcomeObserver.js`, and `exports/character-dag-drama-framework-robustness-`

A stricter follow-up then targeted the failure mode exposed by the stronger controls: compressed state could previously answer transfer scenes with generic "some condition must hold" language. The observer was tightened so transfer closure requires naming the public prior check carried by matched character state, while compressed and stale controls withhold that concrete check. A larger real generated-learner `state_dependent_transfer` matrix then ran two seeds across four fixture families (`base`, `ratio_series`, `definition_boundary`, `causal_identification`), seven arms, and eight scenes ($N = 16$/arm/family). All four family runs pass acceptance gates under `character-dag-drama-observer.v0.5`. Full first-response success is 15/16, 14/16, 14/16, and 13/16; policy-only is 6/16, 3/16, 5/16, and 4/16; shuffled-state is 7/16, 6/16, 6/16, and 5/16. The transfer-specific contrast is sharper: full closes 6/6, 5/6, 6/6, and 5/6 transfer scenes on the first response, while policy-only, shuffled, stale, overconfident, and compressed controls all close 0/6 transfer scenes in the three non-base families and policy/shuffled remain 0/6 in the base family. Every family records zero target-label leaks and zero public theory/process leaks. This upgrades the Character-DAG drama result from a single-fixture strict-perturbation robustness claim to a fixture-family synthetic robustness claim, but it remains an exploratory file-artifact result, not a Paper 2.0 main mechanism estimate.

The licensed reading is still apparatus-first. The result shows that, inside this synthetic harness, proof-DAG state transitions, resistance-signal routing, peripeteia-form pressure, and an evidence-derived character state can be coordinated as one adaptation policy layer, and that the matched-state route separates from a shuffled-state control on transfer. It does **not** show human learning, real interior change, deployed tutoring reliability, or a general "adaptive subject" capacity. It licenses the next development question: whether the same framework can be run on larger, preregistered synthetic families and then translated into human-learner pilot hypotheses without treating synthetic character-development form as learning evidence.

Status (per §5.12.6). Post-hoc exploratory mechanism probes; synthetic learner throughout; no human outcomes; no alpha correction; policy realization remains programmatic; learner generation is real-LLM-backed only in the `learner-mode=llm` runs. Evidence is file-based rather than DB-backed. For the longitudinal state probe, `exports/adaptive-dag-resistance-character-development-llm-real/report.md` and `summary.json` are the canonical record. For the Character-DAG drama framework, `exports/character-dag-drama-framework-transfer-specificity-real-family/robustness-report.md` and `robustness-summary.json` are the strongest current strict robustness record; `exports/character-dag-drama-framework-robustness-policy-repair-real/`, `exports/character-dag-dra` `exports/character-dag-drama-framework/`, and `exports/character-dag-drama-framework-llm-mock/` remain earlier contrast/smoke artifacts. The results support only the modest efficiency/apparatus claims stated above.

6.9 Evidence-Bound Hypothesis Tracking: An Orthogonal Extension to the State-Policy Architecture (A14)

This subsection reports a separate architectural-extension arc, run on the same LangGraph runner as §6.8 but addressing a different gap. §6.8 asked whether an externalised `learnerProfile` plus a programmatic policy graph produces measurable strategy shifts on a trap-scenario instrument. §6.9 asks a discipline question, not a behaviour question: *when the tutor maintains a model of the learner, can the controller force every claim in that* *model to be cited against verbatim evidence from the dialogue?* The §6.8 `learnerProfile` is a single object whose fields ("the learner has misconception X", "confidence is 0.4") are written each turn without an audit trail — there is no way, from the persisted trace, to ask *what learner utterance the tutor's claim about misconception X is grounded in.* The §6.8.6 state-richness reversal is one consequence of that opacity: a richer profile did not improve strategy selection partly because the apparatus could not tell whether the additional fields were tracking the learner or accumulating tutor speculation. §6.9 is the first stage of a longer-horizon answer (the A14 arc in `TODO.md`): replace the unstructured profile-write with an append-only observation ledger and a typed-hypothesis layer whose claims must reference ledger entries by id.

The arc reaches a stable resting point at the end of this subsection: three graph nodes (extractor, updater, validator) are implemented and verified on a small ($N = 3$ scenario) real-LLM smoke; a paired ablation between `cell_126` (updater only) and `cell_127` (updater plus validator) on the two scenarios that both cells ran cleanly measures the validator-enabled architecture's marginal contribution; and a Stage 5 pedagogical-outcome scoring of all three evidence-bound cells at pooled $N \approx 33$ against §6.8's binary and graded instruments (§6.9.7) closes the arc. Per §5.12.6, what follows is *post-hoc, exploratory, and infrastructural*: no pre-registration, no hypothesis test, with the §6.8-vs-§6.9 comparison added at Stage 5 (§6.9.7) as a post-hoc exploratory contrast. The contribution is mechanism availability — the apparatus needed to ask grounding-discipline questions in later work — together with one aggregate mechanism signal in hand (promotion appears only in the validator-enabled arm on the matched subset, while a later trace-integrity audit shows that exact per-decision attribution is not recoverable from the legacy snapshots; §6.9.5–§6.9.6) and a Stage 5 pedagogical-outcome reading (§6.9.7) that is negative-leaning on the apparatus as a whole: the audit chain does not pay in §6.8's strategy-shift coin, and applied to §6.8.6's best configuration it is net-negative on both the binary and graded channels. §6.9.8 then steps outside A14 specifically to close the broader §6.8.8/§6.9 question across the *realisations* of a `state → action` policy — implicit forward pass, hand-authored nodes, and a policy *fitted* to the independent outcome — with the one lever the arc had not yet pulled, killed offline before any live run.

6.9.1 Method: Append-Only Ledger, Typed Hypotheses with TTL Two state-schema additions extend the §6.8 `AdaptiveTutorState` (see `services/adaptiveTutor/stateSchema.js`). The first is `evidenceLog` — an append-only ledger reduced by concatenation. Each entry carries an `obs_id` (content-derived sha1 prefix, stable across counterfactual replay branches), the `turn` index, a verbatim `quote` span from the dialogue, a `type` tag drawn from a fixed enum (`learner_self_report`, `learner_action`, `learner_question`, `learner_correction`, `tutor_inference`), and a `validated` boolean set by a substring-match gate (described below). The reducer is append-only so the audit trail cannot be silently rewritten by a later turn.

The second is `hypotheses` — typed tentative claims about the learner, reduced by merge-by-id (last write wins per `hypothesis_id`). Each hypothesis carries a `claim` string, a `confidence` in $[0, 1]$, two arrays of `obs_id` references (`supporting_evidence` and `contradicting_evidence`), a `status` from `{tentative, validated, contradicted, expired}`, and TTL metadata as the pair (`created_at_turn, expires_after_turns`) rather than an absolute `expires_at_turn` — storing the offset keeps creation metadata immutable under counterfactual replay, where turn counters can branch. The merge-by-id reducer is the design's key commitment: a hypothesis revised on turn $t + 1$ overwrites the same entry rather than appending a parallel claim, so the persisted state reflects the controller's current best guess rather than its full history of guesses.

Two new graph nodes operate on these fields. The *evidence extractor* runs each turn after a learner utterance and before the existing profile-update path; it asks the model to return verbatim quotes typed by the enum above. The node then enforces a substring-match gate: each candidate quote is checked against the cumulative dialogue text, and `validated` is set to `true` only if the gate passes (the entry is retained either way, so hallucinated quotes are recorded for audit rather than silently discarded). The *hypothesis updater* runs immediately after the extractor and consumes only the validated subset of the ledger together with the current `hypotheses` list. It is prompted to revise existing entries (re-emit the same `hypothesis_id` when extending support, mark `contradicted` when the new evidence undermines an earlier claim) rather than mint new claims that paraphrase old ones, and to return an empty array when the evidence will not support any tentative inference. The node owns three pieces of bookkeeping the LLM is not trusted with: it derives a stable `hypothesis_id` from the `claim` text when the model leaves the field blank, it filters `supporting_evidence` and `contradicting_evidence` against the set of valid `obs_id`s so the LLM cannot invent references, and it sweeps for TTL expiry (transitions `tentative → expired` when turn > created_at_turn + expires_after_turns). The "node owns bookkeeping; the LLM owns judgment" split is the same pattern §6.7's id-director engine uses to constrain charismatic-persona authorship to a structured envelope.

A new cell `cell_126_state_policy_evidence_bound` registers the extension. Its `adaptive.architecture` is `state_policy_evidence_bound` — the existing `state_policy` graph (§6.8.2) extended with the extractor and updater nodes wired ahead of the profile-update path. All other cells in §6.8 remain unaffected: the schema additions default to empty arrays and the new nodes only run under the `state_policy_evidence_bound` architecture flag.

6.9.2 Stage 2 Verification: 100% Verifiable Quotes, 100% Grounded Hypotheses on Three Trap Scenarios A real-LLM smoke (`scripts/run-adaptive-cell-real-smoke.js`, `claude-code/sonnet` backend, $N = 3$ scenarios × 2 branches = 6 dialogues) tested the extractor (Stage 2a) and the updater (Stage 2b) jointly on three v1 trap scenarios spanning the agency-signal range — `false_confusion_v1` (compliant), `resistance_to_insight_v1` (resistant), `sophistication_upgrade_v1` (questioning).

Stage	Metric	Result	Pre-registered?
2a	Verifiable-quote rate (validated entries / total entries)	128 / 128 = **100.0%**	No — internal calibration target was ≥95%

Stage	Metric	Result	Pre-registered?
2b	Grounded-hypothesis rate (hypotheses with ≥ 1 supporting `obs_id` / total hypotheses)	53 / 53 = **100.0%**	No — internal soft gate was $\geq$70%

Both rates clear their internal calibration targets. Extraction fires across all turns of all three scenarios (per-turn breakdown: every k-of-k on every turn), with no quote failing the substring-match gate. Hypothesis synthesis is also active rather than degenerate: across the three scenarios the updater emits 13, 16, and 24 hypotheses respectively (mean claim length 196–254 characters; status distribution shows 18 of 53 marked `contradicted` rather than `tentative`, so the model uses the enum to retire claims rather than letting them silently rot), and every hypothesis carries at least one valid `obs_id` reference into the ledger. These are the lower bars Stage 2 was designed to clear: *the controller can build a verifiable evidence ledger and the updater consistently cites it.*

The numbers come from a $N = 3$ smoke, not a study; they speak to the implementation passing its own verification gates, not to any pedagogical claim. Generalisation to the full eight-scenario v1 suite or to cross-suite trap stimuli (the §6.8.7 instrument) is not yet measured.

6.9.3 Limitation Surfaced by the Verification: Revision Versus Duplication A separate signal in the same smoke is informative for what Stage 3 has to do but is not itself a Stage 2 result. The merge-by-id reducer collapses two hypothesis records into one only when their `hypothesis_id` strings match. The updater is prompted to re-emit prior turn ids when extending or contradicting an existing claim, and the node derives a stable id from the `claim` text when the LLM omits it. The smoke shows the reducer is rarely firing: on `sophistication_upgrade_v1`, 24 distinct `hypothesis_id`s survive across a 4-turn $\times$ 2-branch dialogue, with `created_at_turn` spread $[0, 0, 1, 1, 1, 1, 2, 2, 2, 3, 3, 3]$ on each branch. That is the same count of survivors as proposals, indicating the model produces fresh claim text on each turn rather than rewriting yesterday's — the content-derived id then differs, and the reducer treats the new claim as a new entity. The grounded-rate gate (Stage 2b's 70% soft target) does not catch this failure mode because it asks "is each claim grounded?", not "is the same claim being maintained across turns?"

This is reported as a limitation, not a result. Three observations bound how it should be read. First, it is what the design predicted Stage 3 would have to take on: the `groundingValidator` node planned for that stage will retain or retire hypotheses against the ledger across turns, which is the natural place to instrument revision-vs-duplication directly. Second, the status discipline observed at Stage 2b (18 of 53 hypotheses `contradicted` rather than `tentative`) is evidence that the model uses the typed-status machinery when prompted to — the duplication is not "the model ignores the schema" but "the model writes new claims when an update prompt would have been a better fit". Third, the apparatus needed to *measure* revision-vs-duplication crisply does not yet exist either: `extractTurnTrace` (`services/adaptiveTutor/persistence.js`) captures terminal `evidenceLog` and `hypotheses` but not per-turn snapshots, so the count of distinct survivors is a proxy for revision rate, not a direct measurement. Closing that proxy gap is logged as Stage 3 work.

6.9.4 What is Defensible; What is Not What §6.9 establishes is narrow and infrastructural. The state schema for an audited learner model — ledger, typed hypotheses, TTL semantics, content-derived ids stable across counterfactual replay — exists and is reducer-correct. Two new graph nodes (`evidenceExtractor` with substring-match validation gate; `hypothesisUpdater` with node-owned bookkeeping for id derivation, evidence-ref filtering, and TTL sweep) operate on a fresh `state_policy_evidence_bound` architecture; both are exercised in mock and real-LLM smokes. On a $N = 3$ real-LLM verification the extractor emits 100% substring-verifiable quotes and the updater emits 100% ledger-grounded hypotheses, clearing the calibration targets the implementation was built to. The contribution is the apparatus, in the same sense §6.8.8 means it: the §6.9 cell adds a layer the apparatus could not previously ask the question about, not a result about whether the layer helps pedagogically.

What §6.9 does not establish is anything about pedagogical outcome (the evidence-bound cell has not been scored on the binary or graded instruments of §6.8, and would need to be at $N \approx$ 24–32 before such a comparison would be informative), nothing about whether the discipline survives a stronger ablation more broadly (varied scenario types, different judge models, longer dialogues), and — as §6.9.5 develops — a single-sided Stage 3 measurement on `cell_127` but no matched cell_126 status-distribution on the same scenarios from which to read the validator's *marginal* contribution beyond what a well-tuned updater alone does. The Stage 3 architecture (`state_policy_evidence_bound_validated`) and its paired cell (`cell_127_state_policy_evidence_bound_validated`) are now implemented — the `groundingValidator` graph node downstream of the updater promotes hypotheses with $\geq$3 supporting `obs_id`s and no contradiction to `validated`, and retires hypotheses with $\geq$1 contradicting `obs_id` and confidence below 0.4 to `contradicted` — and §6.9.5 reports the first real-LLM smoke through that chain. The honest claim across §6.9 remains that a mechanism is becoming *available*, and §6.9.5 adds that the cheap-verifier layer fires non-trivially on the available material; neither claim is about pedagogical payoff.

6.9.5 Stage 3 Result: Validator-Enabled Architecture Reshapes Terminal Status Distribution A real-LLM smoke on `cell_127_state_policy_evidence_bound_validated` (`claude-code/sonnet`, the same $N = 3$ trap scenarios as the Stage 2 smoke — `false_confusion_v1`, `resistance_to_insight_v1`, `sophistication_upgrade_v1` — with `ADAPTIVE_TUTOR_CLI_TRACE=1` for per-call diagnostic visibility) ran the full chain (`evidenceExtractor` → `hypothesisUpdater` → `groundingValidator` → `learnerProfileUpdate` → tutor ego/superego), and yields the following terminal-state distribution across 26 hypotheses synthesized over the three scenarios:

Status	Count	Fraction
`validated`	13	50%
`contradicted`	9	35%
`tentative`	4	15%

Stages 2a and 2b clear at the same rates as the Stage 2 smoke (120/120 = 100% substring-verifiable quotes; 26/26 = 100% ledger-grounded hypotheses), and the revision rate is 20/22 = 90.9%, statistically indistinguishable from the cell_126 post-tune baseline at 92.9%: adding the validator does not break the upstream updater's revision discipline. What the validator-enabled architecture adds is a much more decisive terminal status distribution. Of the 26 hypotheses, 22 (85%) end `validated` or `contradicted` and 4 (15%) remain `tentative`; the 50/35 promote/retire split is consistent with a verifier that is not one-sidedly stamping things `validated`. These are terminal snapshots, not a raw verdict log: because the updater can itself emit status fields and the historical persistence layer collapsed node-level states to one end-of-turn snapshot, 22/26 is a non-tentative terminal-state fraction, not an exact count of validator decisions. Per-scenario the same shape holds: `sophistication_upgrade_v1` (the trap the prior Stage 2 smoke could not complete) lands at 12 hypotheses with 8 validated, 2 contradicted, 2 tentative — the heaviest validation load and also the heaviest revision rate (9/10 = 90%). The architecture-stress contrast carries through.

A latency note speaks to the design intent. Median per-call latencies under `claude-code/sonnet` are: `hypothesisUpdater` ~100s, `learnerProfileUpdate` ~50s, `tutorEgoInitial` ~45s, `groundingValidator` ~18s, with `learnerTurn` / `tutorSuperego` / `evidenceExtractor` all near 20s. The validator's per-call cost is roughly 5× cheaper than the upstream `hypothesisUpdater` it audits, consistent with the cheap-verifier-after-expensive-synthesizer pattern the architecture was built to: high-cost generation feeding low-cost validation, with the validator's verdict-only contract (status transitions only, no rewriting of hypothesis content) holding the budget down.

A paired-ablation run on `cell_126` (validator off; same `claude-code/sonnet` backend; same trap-scenario inputs) measures the marginal contribution. The Stage 2 cell_126 smoke at commit `1e02510` reported only aggregate revision rate, so a re-run was needed to capture the terminal status distribution. Two of the three scenarios completed cleanly; `sophistication_upgrade_v1` crashed on a Zod schema-strictness bug (`evidenceExtractor` emitted `quote: null` for a `tutor_inference` entry; the graph node at `services/adaptiveTutor/graph.js:371` already coerces null quotes to empty strings — the substring-match gate then correctly marks the entry `validated=false` — but the schema rejected the null upstream). Loosened to `quote: z.string().nullable()` with the rationale documented inline. The matched comparison reads off the two scenarios both cells ran cleanly:

Cell	N scenarios	Hypotheses	`validated`	`contradicted`	`tentative`
`cell_126` (updater only)	2	19	0 (0.0%)	5 (26.3%)	14 (73.7%)
`cell_127` (updater + validator)	2	14	5 (35.7%)	7 (50.0%)	2 (14.3%)

Three patterns are visible. (i) **`validated` appears only in the validator-enabled arm on this matched subset.** The updater-only cell produces zero `validated` outcomes across 19 hypotheses, while `cell_127` produces five. This is an aggregate arm-level attribution signal, not proof that each of the five individual transitions was emitted by `groundingValidator`: the legacy traces did not persist the raw updater and validator proposals separately. (ii) **`contradicted` is shared across arms.** `cell_126` produces 5 `contradicted` hypotheses; `cell_127` produces 7. The updater itself therefore uses the typed-status machinery to retire claims when prompted, and the two additional retirements in the enabled arm cannot be assigned per decision from the collapsed snapshots. (iii) **Uncertainty drops sharply.** `cell_126`'s 73.7% `tentative` collapses to 14.3% in `cell_127`: enabling the validator is associated with about 59 percentage points less unresolved status (split roughly 35 pp promotion / 24 pp retirement). The clean causal object is the architecture-level contrast — adding the validator node reshapes the distribution — while the exact updater-versus-validator decomposition remains unidentified in these historical traces.

What §6.9.5 *does* establish: the validator-enabled architecture produces terminal verdicts of both kinds at a cost meaningfully lower than the updater it follows, without disturbing the revision discipline upstream; and on the matched two-scenario subset, `validated` appears only when that node is enabled. What it does *not* establish: exact per-decision attribution between updater and validator; any pedagogical-outcome claim (the chain has not been scored on the §6.8 binary or graded instruments, and would need to be at $N \approx 24$–32 before such a comparison would be informative); generalisation beyond the two matched scenarios (`sophistication_upgrade_v1` is the heaviest-load trap in the suite and the enabled architecture's contribution there is unmeasured at the paired level — only the single-sided cell_127 reading at 8 `validated` / 2 `contradicted` / 2 `tentative` is in hand); robustness across judge models or generation backends; and whether the terminal distribution would survive scenarios where the upstream updater is already well-tuned to emit `validated` directly. The defensible mechanism claim is arm-level: adding a cheap verifier downstream of an expensive synthesizer measurably reshapes the terminal status distribution on this material.

A later zero-model trace-integrity audit (2026-08-04) makes that attribution boundary exact. Across the completed Stage-5 evidence-bound corpus (`cell_127`, 33 dialogues; `cell_128`, 34; 134 original/counterfactual branches), the persisted end-of-turn snapshots contain 493 observed terminal events: 268 hypotheses first appear in a recorded turn already `validated` or `contradicted`, while 225 change to a terminal status after a previously observed status. All 493 carry non-empty status-relevant evidence references that exist in the validated ledger at that turn; 492/493 (99.8%) also satisfy the corresponding prompt threshold. The single exception is a `cell_127` counterfactual transition to `validated` at confidence 0.7 with six supporting *and two contradicting* ledger references, violating the no-contradiction promotion criterion. This is strong snapshot-level structural consistency, but it is not a silent-drop rate or a reasoning-citation audit: unknown `hypothesis_id` proposals were discarded before persistence, raw `reasoning` was not stored, and the updater-versus-validator intermediate state was collapsed. The exact historical silent-drop rate is therefore **not identifiable**, not 0%. Prospective `grounding_validator_call_audit` and `grounding_validator_decision_audit` events now record every proposal, application/drop reason, cited ledger id, and threshold check without changing the policy. An eight-dialogue deterministic mock smoke (16 branches, 66 validator calls, 31 decisions) verifies the instrumentation at 0/31 dropped and 31/31 structurally grounded; those mock rates validate the apparatus only and are not estimates of real-model behaviour.

A methods note resolves a separate open item (`TODO.md` task #34) raised by the Stage 2b smoke. That earlier run was reported as "hanging silently for 48 minutes" on the third trap scenario. With per-call CLI tracing now on, the cell_127 smoke ran the same scenario set to completion in ~100 minutes wallclock without any individual role-call exceeding the 6-minute CLI subprocess timeout; the longest single call was 134s (`hypothesisUpdater` on `sophistication_upgrade_v1`). The prior report was a misdiagnosis. The earlier smoke was not hung but slow: many sub-6-minute calls compounded into a wallclock window that exceeded the operator's patience before any of them tripped the existing timeout. The instrumentation added in `services/adaptiveTutor/realLLM.js` (env-gated `ADAPTIVE_TUTOR_CLI_TRACE=1` start/done lines per CLI call, plus an unconditional stderr line on timeout-fire with role, elapsed, output-bytes, and first-byte timing) closes the diagnostic gap that produced the silent-wait failure mode. Future hangs surface a role + duration + stream-state signal in real time rather than after the fact.

6.9.6 Stage 3 Synthesis: The Audit Chain Works; the Pedagogical Question Is Settled in §6.9.7 Stepping back from the cell-by-cell results to the discipline question §6.9 opened with — when the tutor maintains a model of the learner, can the controller force every claim in that model to be cited against verbatim evidence from the dialogue? — the answer is *yes at the persisted hypothesis-snapshot level on three trap scenarios, with every internal gate clearing and a sharp aggregate signal that enabling the validator changes the terminal status distribution; but the legacy traces do not identify the layer that emitted each individual status decision.* Like §6.8.8 and §6.7 before it, the durable result is the apparatus. The later trace-integrity audit in §6.9.5 narrows the earlier asymmetric-attribution reading from a per-decision mechanism finding to an arm-level distributional result and makes exact future attribution prospectively measurable.

What is defensible. Four findings survive their caveats:

1. *The three-stage audit chain holds on real-LLM material at the persisted-state level.* On three trap scenarios (`false_confusion_v1`, `resistance_to_insight_v1`, `sophistication_upgrade_v1`) under `claude-code/sonnet`, Stage 2a clears at 128/128 = 100% substring-verifiable quotes (§6.9.2), Stage 2b at 53/53 = 100% ledger-grounded hypotheses (§6.9.2), and Stage 3 produces a non-trivial terminal distribution (13 `validated` / 9 `contradicted` / 4 `tentative` across 26 hypotheses on `cell_127` at $N = 3$; 85% non-tentative; §6.9.5). The node-owned bookkeeping (id derivation, evidence-ref filtering, TTL sweep) is visible and stable. The historical trace does not, however, preserve the raw judgment emitted by each upstream/downstream LLM role.

2. *Validator-enabled marginal contribution is promotion-heavy at the arm level.* The matched $N = 2$ paired ablation against `cell_126` (validator off; same scenarios, same backend; §6.9.5) reads: `cell_126` produces 0 `validated` / 5 `contradicted` / 14 `tentative` across 19 hypotheses; `cell_127` on the same two scenarios produces 5 `validated` / 7 `contradicted` / 2 `tentative` across 14 hypotheses. Enabling the validator is therefore associated with promotion appearing ($0 \rightarrow 5$), two more retirements ($5 \rightarrow 7$), and ~59 percentage points less tentative state. Because the legacy trace

collapses updater and validator intermediate states, this supports an architecture-level marginal contribution, not wholly attributable individual transitions.

3. *The cheap-verifier-after-expensive-synthesizer pattern is empirically demonstrated.* Median per-call latency under `claude-code/sonnet`: `hypothesisUpdater` ~100s, `groundingValidator` ~18s — the validator costs roughly 5× less per call than the layer it audits (§6.9.5). The verdict-only contract enforced by the Zod schema (LLM owns the status transition only; the graph node preserves `claim`, `supporting_evidence`, `contradicting_evidence`, TTL metadata) keeps the validator's per-call budget tight and prevents it from rewriting hypothesis content. The pattern generalises beyond this specific chain: any agent architecture with a expensive-generator-followed-by-cheap-verifier shape can use the verdict-only contract to bound the verifier's responsibility surface.

4. *Task #34 misdiagnosis resolved via per-call instrumentation.* The Stage 2b smoke's "48-minute silent hang" turned out to be a slow-not-hung condition: many sub-6-minute role-calls compounded into a wallclock that exceeded operator patience without any individual call tripping the existing CLI subprocess timeout. The env-gated `ADAPTIVE_TUTOR_CLI_TRACE=1` start/done log lines plus the unconditional stderr line on timeout-fire (commit `62fa729`) close the diagnostic gap; future hangs surface a role + duration + stream-state signal in real time rather than after the fact (§6.9.5 methods note).

What is not established. Four gaps bound the arc as it closes:

- *Pedagogical-outcome scoring — now run (§6.9.7), and negative-leaning.* As of Stage 5 the chain *has* been scored against §6.8's binary `strict_shift` and 4-dimension graded instruments at pooled $N \approx 33$ on all three evidence-bound cells. The result (§6.9.7): the validator's binary gain over the updater-only `cell_126` (+12 pp) does not survive the graded channel (+0.01 overall — a `cell_123`-style binary/graded divergence in the §6.8.6 sense), and the audit chain applied to §6.8.6's winning `cell_118` minimal profile (`cell_128`) is net-negative on both channels (−10 pp binary, −0.21 graded). §6.9's contribution is therefore mechanism availability + one mechanism finding *plus* a complete negative pedagogical-outcome reading — not an open gap.

- *Generalisation beyond the matched $N = 2$ subset.* The paired-ablation table is anchored on `false_confusion_v1` and `resistance_to_insight_v1`. The third scenario, `sophistication_upgrade_v1`, has only single-sided `cell_127` data (the `cell_126` run on that scenario crashed on a Zod-schema strictness issue, now fixed at `services/adaptiveTutor/realLLM.js:478`). Whether the arm-level status-distribution signal survives the heaviest-load trap in the suite is unmeasured at the paired level.

- *Robustness across judge models, generation backends, or scenarios where the updater already emits `validated` directly.* The internal calibration rates and the paired ablation come from a single backend (`claude-code/sonnet`) and a single scenario family (the v1 trap suite). Whether the validator's verdict-rate would survive a different generation backend, or scenarios where the upstream updater's prompt is already well-tuned enough to emit `validated` unaided, is unmeasured.

- *Whether revision-vs-duplication (§6.9.3's limitation) is improved by the validator.* The Stage 2b smoke surfaced a duplication failure mode (the merge-by-id reducer rarely fires; the updater produces fresh claim text each turn rather than rewriting prior claims). The Stage 3 smoke reports a 90.9% revision rate on `cell_127` versus the 92.9% cell_126 post-tune baseline — statistically indistinguishable, so the validator does not break the upstream updater's revision discipline (§6.9.5), but neither does it visibly improve it. The duplication-vs-revision story is not yet a story; it is a flat reading at the rates the updater was tuned to.

Relation to §6.8.8 and §7.4. §6.8.8 closed on *the apparatus is the contribution; the elaborations mostly are not.* §6.9 fits the same motif: the apparatus surfaces a sharp aggregate asymmetry — promotion appears in the validator-enabled matched arm while the updater-only arm emits zero out of nineteen — and the later audit identifies exactly what the old trace cannot assign at the individual-decision level. Read alongside §6.8 and §6.7, the pattern again rhymes: in all three arcs the durable result is the instrument or typed trace, not the architectural elaboration. Whether the validator-enabled arm-level gain pays in §6.8's adaptive-responsiveness coin is answered in §6.9.7: it does not. The +12-point `strict_shift` gain over `cell_126` has no graded-channel correlate (+0.01 overall), and the full apparatus applied to §6.8.6's best configuration is net-negative on both — structurally disciplined at the persisted-state level but pedagogically inert-to-negative.

Status (per §5.12.6). §6.9 is post-hoc, exploratory, infrastructural, and not pre-registered. No alpha correction applies. This subsection is the *Stage-3* synthesis (three-stage architectural feasibility plus the paired-ablation marginal-contribution measurement); the arc's resting point is Stage 5 (§6.9.7), which adds the pedagogical-outcome scoring at pooled $N \approx 33$ against the §6.8 binary and graded instruments. The cross-reference-not-comparison scope stated here applied to the Stage-3 synthesis as it stood; §6.9.7 offers the §6.8-vs-§6.9 cell contrast explicitly and supersedes this paragraph on that point.

6.9.7 Stage 5 Result: The Validator's Binary Gain Does Not Survive the Graded Channel, and the Audit Chain Degrades §6.8.6's Best Configuration

The §6.9.6 synthesis closed the Stage-3 arc at mechanism availability and named one outstanding measurement: pedagogical-outcome scoring against the §6.8 instruments. Stage 5 runs it. All three evidence-bound cells were scored at pooled $N \approx 33$ on the v1 trap suite (`config/adaptive-trap-scenarios.yaml` — the §6.8.6 instrument) along both §6.8 paths: the binary `strict_shift` (`scripts/analyze-strategy-shift.js`, single-judge `claude-code/sonnet`) and the 4-dimension graded rubric (`scripts/grade-adaptive-dialogue.js`, single-judge GPT-5/codex, grader v1.0) — the identical instruments and judges §6.8.3–§6.8.7 used, so the numbers drop directly alongside the §6.8.6 ablation table. `cell_126_state_policy_evidence_bound` is the evidence-ledger + `hypothesisUpdater` chain with the validator *off*; `cell_127_state_policy_evidence_bound_validated` adds the `groundingValidator`; `cell_128_state_policy_minimal_profile_evidence_bound_validated` runs the same validated chain on `cell_118`'s minimal-profile projection (`{confidence, lastEvidence}`), making it the directly audited counterpart of §6.8.6's winning cell. The two readings prefigured for this stage — (a) does evidence-bound discipline beat the §6.8

baselines; (b) does the validator add a pedagogical-outcome gain over the updater-only cell — are answered, both negative-leaning, and the binary and graded channels diverge exactly as §6.8.4/§6.8.6 warned they can.

Cell	Chain	N	`strict_shift`	family-match	Graded overall (T / E / Q / C)
`cell_126` (`evidence_bound`, validator off)	extractor → updater	33	42.4% (14/33)	90.9%	3.84 (3.76 / 3.39 / 4.18 / 4.03)
`cell_127` (`evidence_bound_validated`)	+ grounding-Validator	33	54.5% (18/33)	93.9%	3.85 (3.76 / 3.52 / 4.12 / 4.00)
`cell_128` (`minimal_profile…_validated`)	validated chain on cell_118's profile	34	58.8% (20/34)	100%	4.13 (4.09 / 3.91 / 4.32 / 4.21)
— §6.8 baselines (same instruments, same judges) —					
`cell_110` (`state_policy`, full; §6.8.3)	—	23	47.8%	87.0%	4.03 (4.26 / 4.30 / 3.96 / 3.61)
`cell_118` (`minimal_profile`, unaudited; §6.8.6)	—	32	68.8%	96.9%	4.34 (4.34 / 4.28 / 4.41 / 4.34)
`cell_124` (`state_policy` cross-suite; §6.8.7)	—	24	62.5%	66.7%	4.22

(T / E / Q / C as in §6.8.6; binary under `claude-code/sonnet`, graded under GPT-5/codex. Within-action refinement is 100% on all three evidence-bound cells — 50 / 44 / 60 same-action turns whose message text changed non-trivially — so the binary fire-rate is measured against genuinely re-authored prose, not a degenerate constant. Counterfactual replay is *active* for these architectures, unlike §6.8.6's no-op P2.2 cells, with counterfactual-divergence 93.9 / 97.0 / 88.2%.)

Reading (b): the validator's binary gain does not survive the graded channel. On the binary instrument the validator pays and the family-discipline signal tightens monotonically: `cell_126` 42.4% → `cell_127` 54.5% (+12.1 pp) → `cell_128` 58.8% (+16.4 pp over `cell_126`), with family-match climbing 90.9 → 93.9 → 100%. That is the pedagogical-outcome correlate the §6.9.6 arm-level status-distribution signal predicted on the channel built to detect *which move, when*; the legacy snapshots do not identify which individual status transitions came from the validator rather than the updater. But the graded channel — the load-bearing direction signal per §6.8.6's template — does not corroborate it for `cell_127`: graded overall is **3.84 → 3.85**, a +0.01 null, no dimension moving more than the +0.13 on `strategy_execution` (itself offset by −0.06 on `strategy_quality` and −0.03 on `pedagogical_coherence`). The validator's +12-point binary gain buys no measurable improvement in the execution, quality, or coherence of the resulting teaching. This is precisely the within-P2.2 pattern §6.8.6 isolated in `cell_123` — top on binary, flat on graded — and the cross-architecture pattern of §6.8.4 (`cell_114`): the binary instrument registers a move-selection change the graded rubric does not see as better teaching. The lone cell that moves *both* needles is `cell_128`: 58.8% binary (top of the three) and 4.13 graded (top, +0.29 over `cell_126`, every dimension up) — structurally the same "only the lean configuration is robust across both instruments" shape §6.8.6 found in `cell_118`.

Reading (a): the audit chain does not reach the §6.8 baselines, and it degrades §6.8.6's best configuration on both channels. The evidence-ledger + updater layer *without* the validator (`cell_126`) sits *below* the plain `state_policy` baseline on both instruments — binary 42.4% vs `cell_110`'s 47.8% (−5.4 pp), graded 3.84 vs 4.03 (−0.19) — so the extractor + updater apparatus is net-negative absent the verifier. Adding the validator lifts `cell_127`/`cell_128` over `cell_110` on binary (54.5 / 58.8% vs 47.8%) but no evidence-bound cell reaches `cell_124`'s 62.5% cross-suite mark or `cell_118`'s 68.8%, and none matches `cell_118`'s 4.34 graded. The sharpest comparison is the direct one: `cell_128` *is* §6.8.6's winning minimal profile with the A14 audit chain bolted on, and it scores **58.8% / 4.13 against the unaudited `cell_118`'s 68.8% / 4.34** — the audit apparatus, applied to the single configuration §6.8.6 found most favourable, costs −10 pp binary and −0.21 graded. The apparatus does not merely fail to unlock §6.8.6's headroom; on the configuration that *had* the headroom, it spends it.

Trust-vs-load resolves to load, on the load-bearing channel. The arc opened (§6.9, §6.9.6) on whether forcing every claim in the learner model to cite verbatim evidence would *unlock* strategy-shift performance (trust) or *tax* it without payoff (load). Stage 5 answers: load, on the graded channel §6.8.6 designated load-bearing. The grounding apparatus is snapshot-consistent and has an arm-level mechanism signal — enabling the validator reshapes the terminal status distribution, although the legacy trace cannot attribute individual transitions to it — but it is pedagogically inert-to-negative: the ledger+updater layer is below the plain baseline; the validator buys a binary-only tightening the graded judge does not read as better teaching; and the whole chain, applied to §6.8.6's best cell, makes it worse on both instruments. This is the §6.8.8 motif once more — *the apparatus is the contribution; the elaboration does not pay* — now with A14's evidence-binding as the elaboration, and read alongside the §6.8.6 state-richness reversal it sharpens that section's lesson: the strategy-shift headroom lives in a *lean* learner profile, and neither enriching it (§6.8.6) nor auditing it (here) recovers it. The §6.9 arc therefore closes mechanism-positive and pedagogical-outcome-negative — a complete finding for the arc in exactly the sense its plan anticipated, not a deferral.

Status and caveats (per §5.12.6). Post-hoc, exploratory, not pre-registered (A14 is an infrastructural arc; Stage 5 was a named deliverable, not a §5.12.2-style pre-registration — no alpha correction across the two readings). Pooled N per cell across run-ids per the §6.8.6 norm: `cell_126` = 33, `cell_127` = 33, `cell_128` = 34 (100 dialogues; 0 grader errors). Single judge per channel — binary `claude-code/sonnet`, graded GPT-5/codex (grader v1.0) — with **no second-judge calibration for the Stage 5 pool**: §5.12.4's Gemini inter-rater check covered cells 110/118/119 only, so the graded numbers here carry the same single-judge GPT-5 caveat §6.8.6 attaches to `cell_120`/`cell_123`. Per §6.8.6's statistical template the binary channel is treated as wide-CI — the +12.1-pp `cell_126`→`cell_127` gap at $N \approx 33$ is the high end of a wide interval (label-permutation $p \approx 0.10$ for a gap of this size at this N, by the §6.8.6 calibration), which is *why* the graded null is the decisive reading rather than the binary lift: the load-bearing channel shows no gain. §5.12.4-limitation-1 (rubric authored non-blind to the cells-110/118/119 binary results) applies only weakly here — the Stage 5 cells post-date the rubric's design, so the graded instrument was fixed before their data existed, a cleaner application than §6.8.6's own. Counterfactual replay is *active* for the evidence-bound architectures (they are in the runner's profile-update set; counterfactual-divergence reported above), unlike §6.8.6's no-op P2.2 cells. The §6.8-vs-§6.9 comparison §6.9.6's Status paragraph declined to offer for the Stage-3 synthesis is offered here explicitly; §6.9.7 supersedes that scope note on that point.

6.9.8 Synthesis: The Learned-Policy Lever — the Last Distinct `state→action` Realisation — Also Does Not Pay (Offline Kill Gate) §6.9.7 closed A14's evidence-bound arc on the §6.8.8 motif. This subsection closes the broader question that motif raises. §6.8 exhibited an *implicit* `state → action` policy (the LLM forward pass conditioned on an externalised learner profile); §6.9.1–§6.9.7 a *hand-authored* one (typed graph nodes); both are mechanism-clean and pedagogically inert-to-negative on the architecture-independent instruments. One realisation had not been tried: a policy *fitted* to the independent outcome from logged data, rather than implicit in the forward pass or written by hand — "the only path on which adaptation would mean what it says", in the closed-loop persona critique's phrase. §6.9.8 takes it under a kill gate pre-registered (`LEARNED-ADAPTATION-PLAN.md`, repo root) to be **offline, zero-API, and read-only on traces already on `main`**, so that a null costs nothing and *completes* the realisation-space synthesis rather than opening a new arc.

Method (offline, ≈ \$0, read-only). `scripts/learned-adaptation-harvest.js` walks the existing `cell_110/118/124/126/127/128` trace files (`logs/tutor-dialogues/`) and the `evaluation_results` DB read-only, emitting one row per trajectory at its pivot decision: context = the A14 typed state at the pre-pivot turn (`evidenceLog/hypotheses/learnerProfile/agencySignal`, learner-profile forward-filled across empty terminal turns); the single bandit action = the pivot-turn `policyAction` family; the reward = the §6.9.7 instruments. 179 of 179 trajectories harvested. The binary label is **fidelity-asserted**: a verbatim copy of §6.9.7's `analyzeBranch` recomputes `strict_shift`, required to equal the canonical `exports/a14-stage5-strategy-shift-finalN.json` per-cell rates to within 10^{-6} or the harvest hard-aborts. They match exactly (`cell_126` 0.4242, `cell_127` 0.5455, `cell_128` 0.5882; $\Delta = 0$), and the recovered baseline rates reproduce §6.9.7's table to the decimal (`cell_110` 47.8%, `cell_118` 68.8%, `cell_124` 62.5%) — the label scored here is provably the §6.9.7 instrument, not a re-implementation that could drift toward a self-flattering result. `scripts/learned-adaptation-policy-ope.py` (numpy only, no dependence on any controller-internal metric) fits the simplest policy — multinomial logistic over state features for the binary channel, ridge-linear for the graded channel — under **leave-one-`scenario_type`-out grouped cross-validation** (13 trap families; the policy must select from state, not memorise family identity), evaluated off-policy two ways: a Direct Method (fitted reward model) and an assumption-light on-policy-agreement value (empirical reward on the subset where the logged action family already equals the policy's argmax). Confidence intervals are **cluster bootstrap by `scenario_type`** ($B = 2000$, seed `20260517`) — the honest variance unit, since trajectory outcomes vary overwhelmingly *between* trap families and a row-level bootstrap would manufacture a false pass.

The pre-registered kill gate (frozen in the plan before the OPE was computed): the fitted policy must beat **both** the implicit baseline `cell_110` *and* A14's hand-authored updater `cell_126`, lower 95% cluster-bootstrap CI of the difference > 0, on the primary binary `strict_shift` label. Beating only the hand-authored cell is not a result — it is the §6.8.8 motif ("structured elaboration underperforms the strong implicit base") reappearing, not escaped.

Off-policy contrast	vs `cell_110` (implicit base)	vs `cell_126` (A14 hand-authored)
Direct Method Δ, binary `strict_shift` (primary)	+9.0 pp, 95% CI $[-0.9, +18.0]$ — does **not** beat	+14.4 pp, CI $[+4.8, +23.0]$ — beats
Graded-channel Δ (corroborating)	-0.03, CI $[-0.16, +0.13]$ — does not beat	+0.16, CI $[+0.03, +0.33]$ — beats

The assumption-light cross-check agrees with the Direct Method against the implicit baseline: on the $n = 95$ agreement subset the fitted policy's empirical value is 0.326 binary / 3.88 graded, *below* `cell_110`'s 0.478 / 4.03 on both. **The gate fails.** The fitted policy is statistically indistinguishable from the implicit in-context baseline on the load-bearing label and clears only the *weakest* A14 cell — on both channels, in agreement across two independent estimators.

Reading. This is the structural prediction of the project's own adaptivity synthesis, now measured rather than argued: the A14 typed state largely *re-encodes* the scenario family the strong base model already conditions on in-context, so a `state → action` policy tested *off-family* (the grouped-CV requirement) has little headroom over the forward pass it is trying to improve. A fitted policy that matches the implicit base while beating only the thinnest hand-authored node is §6.8.6's "only a lean configuration is robust" and §6.8.8's "the apparatus is the contribution; the elaboration does not pay", one level up: here the elaboration is *learning* the policy rather than writing or enriching it, and it does not pay either. In learning-analytics terms the fitted parametric `state → action` policy evaluated here is the *learned*-policy realisation of a learner-state backbone in the lineage of knowledge tracing (Piech et al., 2015; Scarlatos et al., 2025) — not a full mastery-posterior, learning-outcome-validated knowledge-tracing design, a distinction the null respects rather than erases; it localises where that learned learner-state backbone meets a strong in-context base.

The non-parametric sibling, now run (not argued). A15's planned cross-dialogue *retrieval* is the non-parametric, instance-based realisation of the same fitted `state → action` map: rather than a parametric logistic, embed the pivot state, retrieve the top-k nearest past `(state, action, outcome)` tuples, and select the action family that *worked* among them. `scripts/retrieval-adaptation-knn-ope.py` runs it on the **identical** harvest table, state features, Direct-Method reward model, grouped leave-one-`scenario_type`-out CV, and cluster bootstrap as the parametric gate above — the only substantive change is the policy form ($k = 5$ pre-declared; a $k \in \{3, 5, 10, 15, 25\}$ sweep reported as sensitivity, not selection). The Direct Method *appears* to clear the gate (binary value 0.749, beating both baselines at every k), but that is a **base-rate exploit, not transferable adaptation**, and three pre-registered controls (`scripts/retrieval-adaptation-knn-diagnostics.py`) establish it: (i) the minority action families carry high marginal `strict_shift` *by the label's construction* (`diagnostic` 0.96, `repair_affective` 0.92, `scaffolding` 0.89 vs the `substantive_engagement` default's 0.30 — a rare family is "correct" precisely on the scenarios that expected it), so a reward-greedy selector that merely tilts toward rare families scores high on the model-based estimator without teaching better; (ii) a label-permutation placebo confirms the lift is not pure noise (real DM 0.749 above the shuffled-label null band centred at 0.564), but a *behaviour-cloning* k-NN that retrieves similar states and copies what was *done* rather than what *worked* falls to 0.450, *below* baseline — the apparent gain requires reward-greediness, not retrieval; and decisively (iii) on the **transfer-valid on-policy-agreement channel** — the assumption-light estimator using the actual held-out outcomes of the policy's own picks, the only one identified without logged propensities for a policy that deviates from the logged action on \$ \$82% of decisions — the retrieval policy is **0.3125, below both baselines (0.478, 0.424)**, statistically indistinguishable from the parametric logistic's 0.326. The neighbour-reward signal does not cross the leave-one-family-out boundary in either policy form; the gate fails on the channel that is identified.

Disposition. The realisations of a `state → action` policy that have been built and scored on the architecture-independent instruments — implicit (§6.8), hand-authored (§6.9.7), fitted-parametric (§6.9.8), and now fitted-non-parametric (the retrieval sibling, above) — are all null relative to the implicit base on the transfer-valid channel; A15's nearest-neighbour *retrieval* is thus **run and null, not merely designed and gated**. With the learned lever pulled and null the §6.8.8/§6.9.7 motif is **exhaustive over the policy realisations actually tried, not merely repeated**. This is the fourth *in-paper* convergent negative on adaptive elaboration (§6.7 id-director, §6.8.8 state-richness/bilateral-ToM, §6.9.7 A14 evidence-binding, §6.9.8 here); the archived closed-loop persona prototype is a fifth that lies outside the paper's claim set, named for completeness only. Per the plan's frozen kill criteria the arc closes here: both offline gates failed for free, so there is **no live confirmation run, no counterfactual stage, and no tuning retry** — a clean null *is* the closing result. Artifacts: `exports/learned-adaptation-table.csv`, `exports/learned-adaptation-{table,ope}.meta.json`, `exports/retrieval-adaptation-knn-{ope,diagnos`

Status and caveats (per §5.12.6). Post-hoc and exploratory in the §6.9 sense, with one twist that strengthens it: the *kill gate* was pre-registered before the OPE was computed and was deliberately constructed to be offline and zero-API so that the decision rule could not be revised after seeing a live number — there is no live number. No alpha correction (one pre-declared primary contrast; the graded channel corroborates, it is not a second test). The off-policy estimate is identified by a Direct Method plus an assumption-light on-policy-agreement value rather than importance sampling, because the logging policy (an LLM forward pass / a hand-authored node) carries no recorded propensities; the two estimators agree on the decisive comparison (indistinguishable from `cell_110`), which is the conservative reading and the one the gate rules on. The binary label's identity with the §6.9.7 instrument is asserted by exact recomputation ($\Delta < 10^{-6}$), closing the closed-loop-scoring failure mode by construction. Single judge per channel, inherited unchanged from §6.9.7 (binary `claude-code/sonnet`; graded GPT-5/codex grader v1.0) — this analysis introduces no new judging, only a policy fitted to already-scored outcomes. The claim is bounded: the *learned* lever does not beat the implicit base on the v1-trap-suite instruments, not that no fitted policy could help on any instrument. The non-parametric retrieval gate inherits this bound and adds one of its own: it tests retrieval's effect on the *single pivotal* `state → action` decision (the harvest unit), the necessary condition for the multi-turn slope-improvement A15 targeted — if retrieval cannot beat the base on the one decision it keys on, the slope assembled from such decisions will not improve — but a *direct* turn-by-turn slope test would require the live cell, which this offline null does not justify building (that build, with its embedding cache and paid confirmation, is exactly the "add embeddings and rerun" treadmill the arc's anti-thrash discipline forbids; cf. §6.10).

6.10 Modelling the Concealed Interior — the *Signal* Axis Also Does Not Pay (Offline Kill Gate)

§6.9.8 closed the `state → action` arc over its *policy realisations* (implicit, hand-authored, fitted). It left one axis untested. Every in-paper negative on adaptive elaboration — §6.7 (id-director persona), §6.8.8 (state-richness / bilateral-ToM), §6.9.7 (A14 evidence-binding), §6.9.8 (fitted policy) — shares one shape: each *re-encoded what the strong base model already infers in-context from the surface transcript.* The arc's one *apparent* exception was a disposable mechanism-triage prototype outside this paper's claim set (`prototypes/adversarial-superego-mvp/`; §6.3.10): an adversarial superego with edit rights over the ego's system prompt reliably reached the correct trap-repair move where ignorable advice did so only intermittently. The paper put exactly that mechanism to its own pre-registered test (A16, §6.3.10) on the externalised `policyAction` instrument: it produced a *level/reliability* effect, **not** the per-turn *rate* that would escape §6.3, and the pre-registered primary nulled (S1−S0 $d = -0.167$). The arc therefore carries **no in-claim-set positive** on an architecture-independent channel. The dividing principle the convergent negatives cohere around — gains require signal the base does **not** already read in-context, not a richer restatement of what it does — is therefore the *theoretical* reading that renders the nulls intelligible, not a datapoint the arc itself licenses. This subsection tests the one signal-axis lever the arc had not pulled, under a kill gate pre-registered (`ADAPTATION-PLAN-2.0.md`, repo root; rationale in `ADAPTATION-2.0-DIALOGUE.md`) — like §6.9.8 — to be **offline, zero-API, and read-only on traces already on `main`**, so a null *completes* the synthesis at ≈ $0 rather than opening a new arc.

The candidate unread signal. The RLHF base reads the learner *at face value* (the sycophancy / agreeableness bias). The v1 trap suite is *built from* concealment cases (`polite_false_mastery`, `false_confusion`, `affective_shutdown`, `epistemic_resistance`, `answer_seeking_to_productive_struggle`) — exactly the cases where the truthful reading diverges from the surface utterance. And this study **uniquely owns the ground truth**: the learner is a full ego–superego agent (§3); only `learner/final_output` is externalised, while the hidden `learner_ego_initial / learner_superego / learner_ego_revision` deliberation is in the logged trace and the tutor never sees it. The proposed mechanism (a per-turn, ephemeral *concealment-inference* step generating the tutor move against an inferred latent state, scored by an independent rubric-critic) is only worth building if that manifest≠latent discrepancy is a *separable, structured* signal the surface does not already determine. That necessary condition is **offline-decidable for free** against the real hidden trace — no mechanism need be run to falsify its precondition.

Pre-registered probe (`scripts/adaptation2-stage0.py`, frozen before running; numpy + stdlib only). Unit: one learner turn with the full hidden triplet plus its externalised output. Target (the manifest≠latent quantity): 1 − tfidf-cosine(*ego_initial*, *final_output*), the magnitude by which the public utterance departs from the private first reaction; a token-Jaccard variant is the frozen robustness target. Predictor set **A** "surface-only" = features computable from the *manifest* dialogue alone (preceding tutor message + learner final output + structure) — exactly what the base conditions on, no hidden text. Predictor set **B** "latent-cause" = features from the `learner_superego` critique text *only* (excludes `ego_initial` and `ego_revision`, so it cannot trivially reconstruct the target). Estimator: ridge ($\lambda = 1.0$), leave-one-`scenario_type`-out grouped CV, cluster bootstrap by `scenario_type` ($B = 2000$, seed `20260517`) — the honest variance unit, since outcome variance is overwhelmingly between scenario families. **Frozen P1 pass (both required): lower 95% CI of $R^2_B > 0$ (the target is structured, a legible latent cause exists) *and* lower 95% CI of $(R^2_B - R^2_A) > 0$ (the surface misses structure the latent cause explains).**

Result. Harvest $n = 1604$ triplet-turns, **9 scenario-type families** (`epistemic_resistance_impasse`, `affective_shutdown_impasse`, `mood_frustration_to_breakthrough`, `productive_deadlock_impasse`, `mutual_transformation_journey`, `misconception_correction_flow`, and three trajectory families), 11 profiles, 2589 dialogues. On the primary target:

channel	grouped-CV R^2	95% cluster-bootstrap CI
A surface-only (= what the base reads)	+0.086	—
B latent-cause (`learner_superego` text only)	−0.026	[−0.212, −0.032]

channel	grouped-CV R^2	95% cluster-bootstrap CI
gap $B - A$	-0.111	$[-0.234,\ -0.010]$

Both frozen conditions are violated, and not marginally: the latent-cause channel carries **no recoverable out-of-group structure** (R^2_B negative, CI entirely below 0) and is **significantly worse than the surface** (gap CI entirely below 0). The frozen robustness target (token-Jaccard) concurs ($R^2_A = +0.039$, $R^2_B = -0.022$, CI $[-0.206,\ -0.018]$). **P1 fails.**

Reading — why a pervasive phenomenon is nonetheless not a usable signal. The null is *not* an absence of concealment: the target is not floored (mean cosine distance 0.647, sd 0.101; 99.6% of turns in the graded interval, near-zero only 0.3%) — the simulated learner's public utterance departs substantially from its private first reaction almost every turn. The decisive quantity is the variance decomposition: only 4.3% of the concealment-gap variance is *between* the nine trap families (per-family means 0.606–0.677, a 0.07 spread) and 95.7% is idiosyncratic *within* family. The gap does not track the trap type, is barely recoverable from the surface ($R^2_A \approx 0.09$), and is *not* recoverable from the logged hidden deliberation at all ($R^2_B \approx 0$; corr(*target*, len *superego*) ≈ -0.14 — the superego critique is of roughly constant length and register regardless of how much the learner will actually revise). The premise of the signal-axis lever was that the owned hidden trace is a *privileged latent variable* the tutor lacks. The data falsify that premise: the "hidden" deliberation is **itself another stream of in-context LLM text with the same surface character**, and it does not encode — in any recoverable, family-transferable form — the very concealment it was theorised to explain. This is the unifying lesson of §6.7–§6.10 stated at its sharpest: there is no privileged structured interior beneath the text to model, *not even when the apparatus generates and logs that interior itself.* This is consistent with the machine-theory-of-mind literature, in which LLM mental-state tracking degrades under interaction stress (Kim et al., 2023) and under trivial perturbations of classic false-belief tasks (Ullman, 2023).

Symbolic corroboration (out of claim set). A later follow-up reaches the same null from the opposite methodological direction. A *symbolic* realisation of the concealed-interior lever — a per-turn description-logic theory-of-mind ABox accumulated by the tutor, strictly more inspectable than this section's lexical proxy because an EL consistency check can interrogate it directly — was supplied to the tutor as live guidance and tested in a small real-backend multi-turn A/B (three stress scenarios, one run each). It is null over plain ontology guidance: the consistency check fires correctly — it flags surface compliance, a learner claiming the conclusion while signalling weak ownership, as a *grounded inconsistency* — yet only re-encodes a move the ontology guidance already prompts, and the extra directiveness if anything lowers the learner outcome. The no-privileged-interior reading therefore survives the shift from a free-text interior to a formal, checkable one. This is an exploratory screen (single backend, one run per scenario, heuristic outcome detector), out of the paper's claim set and named for completeness only; write-up and artifacts in `notes/ontology/` (`scripts/run-ontology-tom-ab.js`).

Disposition. Per the plan's frozen kill rule, executed without relitigation: **no Stage 1, no live run, no counterfactual stage, no tuning retry** — the offline null *is* the closing result. With §6.7 / §6.8.8 / §6.9.7 / §6.9.8 this is the **fifth in-paper convergent negative** on adaptive elaboration (the archived closed-loop persona prototype is a sixth outside the paper's claim set, named for completeness only), and the first to test the *signal* rather than the *policy form*: §6.9.8 exhausted the policy-realisation axis; §6.10 closes the signal axis. The adaptation arc is closed on both. The Weber companion probe (P2: tutor charisma → learner-receptivity *uptake*) was **structurally unmeasurable** on the existing corpus: the id-director charisma cells (100–109; §6.7) were run with scripted `unified` learners and scored tutor-only ($\leq$ 38/505 rows carry any learner-side score, 0 carry learner interiority or a receptivity index). That corpus gap is itself the concrete answer to why the charisma lever was never realised — the §6.7 arc instrumented the *performance* of charisma, never its *uptake* (Weber's legitimacy as *conferred by the other*, not performed). Artifacts: `exports/adaptation2-stage0-table.csv`, `exports/adaptation2-stage0-probe.meta.json`.

Status and caveats (per §5.12.6). Post-hoc and exploratory in the §6.9 sense, with the §6.9.8 twist that strengthens it: the *kill gate* was pre-registered before the probe was computed and was constructed offline and zero-API so the decision rule could not be revised after seeing a number. No alpha correction (one pre-declared primary contrast; the robustness target corroborates, it is not a second test). The offline target is *lexical* (tf-idf cosine / token-Jaccard), a noisy proxy for *semantic* concealment; this bounds the claim to lexically-separable concealment structure and is stated, not hidden. It does not reopen the gate, for two pre-registered reasons: the variance decomposition (family-invariant in mean, idiosyncratic within) is a property of the distribution that holds regardless of proxy fidelity; and the proxy plus kill rule were frozen *precisely* to forbid the "add embeddings and rerun" treadmill that the arc's anti-thrash discipline (§6.9.8) exists to stop. No new judging — the probe consumes only already-logged traces. The claim is bounded: modelling the concealed interior buys no separable structure over the surface *on the v1-trap-suite logged ego–superego traces*, not that no interiority model could help on any instrument; that is the close of this arc, not a universal impossibility theorem.

Neighbouring modules: Module 3 — The Tutor Did Not Become More Adaptive; Module 4 — How We Checked Every Claim.

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